3_10xHuman_8CLC_comparison_wSTAR

August 31, 2026

[1]:
```
library(Seurat)
library(ggplot2)
library(pheatmap)
library(ggpubr)
getwd()
dir.create("figures_10xHuman_8CLC")
dir.create("data")


colorAge = c("old"="#7a646a",
             "young"="#88bce7")

colorTools = c("SoloTE"="#A4DD9B",
               "Stellarscope"="#4f5d93",
               "STARsolo"="#f0df93")

dataset_id <- "10xHuman_8CLC"

library(ggbeeswarm)
```

```
Loading required package: SeuratObject


Loading required package: sp

'SeuratObject' was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall 'SeuratObject' as the ABI
for R may have changed

'SeuratObject' was built with package 'Matrix' 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall 'SeuratObject' as
the ABI for 'Matrix' may have changed


Attaching package: 'SeuratObject'


The following objects are masked from 'package:base':
```

```
intersect, t
```

'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

```
Warning message in dir.create("figures_10xHuman_8CLC"):
"'figures_10xHuman_8CLC' already exists"
Warning message in dir.create("data"):
"'data' already exists"
```

[2]:
```
theme_paper <- function(base_size = 17, base_family = "") {
  theme_minimal(base_size = base_size, base_family = base_family) +
    theme(
      plot.title = element_text(face = "bold", size = base_size + 2, hjust = 0.
↪5),
      axis.title = element_text(size = base_size),
      axis.text  = element_text(size = base_size * 0.9),
      legend.title = element_text(size = base_size),
      legend.text  = element_text(size = base_size * 0.9),
      panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
      plot.margin = margin(10, 10, 10, 10)
    )
}
theme_set(theme_paper())
```

1 Pre

[3]:
```
# Import Stellarscope object
objTE_stellarscope <- readRDS(paste0("data_",dataset_id,"/
↪stellarscope_",dataset_id,"_seuratObj.RDS"))
# Import SoloTE object
objTE_soloTE <- readRDS(paste0("data_",dataset_id,"/
↪soloTE_",dataset_id,"_seuratObj.RDS"))
# Import STARsolo object
objTE_STARsolo <- readRDS(paste0("data_",dataset_id,"/
↪STARsolo_",dataset_id,"_seuratObj.RDS"))
```

[4]:
```
objTE_stellarscope
objTE_soloTE
objTE_STARsolo
```

```
An object of class Seurat
234752 features across 125 samples within 1 assay
Active assay: RNA (234752 features, 4000 variable features)
```

```
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap
```

```
An object of class Seurat
365333 features across 125 samples within 1 assay
Active assay: RNA (365333 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap
```

```
An object of class Seurat
224828 features across 125 samples within 1 assay
Active assay: RNA (224828 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap
```

[5]:
```
conversionTable <- read.table("annotation/annotation_hg38_conversion_withAge.
↪tsv") # generated with annotation_scripts/create_annotations_hg38.Rmd
head(conversionTable)
```

[ ]:
```
# get TEs in common between Stellarscope and SoloTE

stellarscopeTEs <- Features(objTE_stellarscope)

table(stellarscopeTEs %in% conversionTable$stellarscopeID)

soloTEs <- Features(objTE_soloTE)
# remove "SoloTE" from the name of the TEs and subtitute "|" and "_" with "-"
soloTEs <- gsub("\\|", "-", soloTEs)
soloTEs <- gsub("\\_", "-", soloTEs)
soloTEs <- gsub("\\?", "", soloTEs)

table(soloTEs %in% conversionTable$soloteID)

TEsInStellarscope_SoloTE <- intersect(␣
↪conversionTable$locusID[conversionTable$stellarscopeID %in% stellarscopeTEs],
                        conversionTable$locusID[conversionTable$soloteID %in%␣
↪soloTEs] )
length(TEsInStellarscope_SoloTE)
```

```
  TRUE
234752
```

```
 FALSE   TRUE
   672 364661
```

212848

```
[ ]: # get TEs in common between Stellarscope and STARsolo

stellarscopeTEs <- Features(objTE_stellarscope)

table(stellarscopeTEs %in% conversionTable$stellarscopeID)

STARsoloTEs <- Features(objTE_STARsolo)


table(STARsoloTEs %in% conversionTable$stellarscopeID)

TEsInStellarscope_STARsolo <- intersect(␣
 ↪conversionTable$locusID[conversionTable$stellarscopeID %in% stellarscopeTEs],
                        conversionTable$locusID[conversionTable$stellarscopeID␣
 ↪%in% STARsoloTEs] )
length(TEsInStellarscope_STARsolo)
```

```
  TRUE
234752
```

```
  TRUE
224828
```

165330

```
[ ]: STARsoloTEs <- Features(objTE_STARsolo)
table(STARsoloTEs %in% conversionTable$stellarscopeID)

TEsInAll <- intersect( TEsInStellarscope_SoloTE,
                       conversionTable$locusID[conversionTable$stellarscopeID␣
 ↪%in% STARsoloTEs] )

length(TEsInAll)
```

```
  TRUE
224828
```

151286

```
[ ]: commonCellsAll <- intersect(intersect(Cells(objTE_stellarscope),␣
 ↪Cells(objTE_soloTE)), Cells(objTE_STARsolo))
length(commonCellsAll)
```

125

[ ]:
```
# # locus id of common TEs
tes_soloTE_conv <- conversionTable$locusID[match(Features(objTE_soloTE),␣
↪conversionTable$soloteID)]
tes_stellarscope_conv <-␣
↪conversionTable$locusID[match(Features(objTE_stellarscope),␣
↪conversionTable$stellarscopeID)]
tes_starsolo_conv <- conversionTable$locusID[match(Features(objTE_STARsolo),␣
↪conversionTable$stellarscopeID)]

commonTEs_conv <- intersect(intersect(tes_soloTE_conv, tes_stellarscope_conv),␣
↪tes_starsolo_conv)
commonTEs_conv <- commonTEs_conv[!is.na(commonTEs_conv)]
length(commonTEs_conv)
```

151286

[ ]:
```
# ALL commmon TEs
# reconvert to original name
common_soloTE <- conversionTable$soloteID[match(commonTEs_conv,␣
↪conversionTable$locusID)]
common_stellarscope <- conversionTable$stellarscopeID[match(commonTEs_conv,␣
↪conversionTable$locusID)]
common_STARsolo <- conversionTable$stellarscopeID[match(commonTEs_conv,␣
↪conversionTable$locusID)]

# extract matrix of common TEs
stellarscopeMatSub <- GetAssayData(objTE_stellarscope[common_stellarscope,␣
↪commonCellsAll], layer = "data")
setdiff(common_stellarscope, rownames(stellarscopeMatSub))
stellarscopeMatSub <- stellarscopeMatSub[common_stellarscope, commonCellsAll] #␣
↪to reorder

STARsoloMatSub <- GetAssayData(objTE_STARsolo[common_stellarscope,␣
↪commonCellsAll], layer = "data")
STARsoloMatSub <- STARsoloMatSub[common_stellarscope, commonCellsAll] # to␣
↪reorder

soloTEMatSub <- GetAssayData(objTE_soloTE[common_soloTE, commonCellsAll], layer␣
↪= "data")
soloTEMatSub <- soloTEMatSub[common_soloTE, commonCellsAll] # to reorder

identical(colnames(stellarscopeMatSub), colnames(soloTEMatSub))
identical(colnames(STARsoloMatSub), colnames(soloTEMatSub))

matSubs <- list()
matSubs[["Stellarscope"]] <- stellarscopeMatSub
matSubs[["SoloTE"]] <- soloTEMatSub
```

```
matSubs[["STARsolo"]] <- STARsoloMatSub

gc()
```

TRUE

TRUE

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	27161432	1450.6	64011664	3418.6	64011664	3418.6
Vcells	241791656	1844.8	464954650	3547.4	464944549	3547.3

[ ]:
```
# matrices with common cells and TEs
mat_stellarscope <- as.matrix(stellarscopeMatSub)

mat_soloTE <- as.matrix(soloTEMatSub)

mat_STARsolo <- as.matrix(STARsoloMatSub)


gc()
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	27161823	1450.6	64011664	3418.6	64011664	3418.6
Vcells	298524604	2277.6	464954650	3547.4	464944549	3547.3

2 Correlations

[ ]:
```
avgExprCommonTEs <- rowMeans(mat_stellarscope)
# compute average expression in the cells where each TE is expressed
sumExprCommonTEs <- rowSums(mat_stellarscope)
nCellsExprTEs <- rowSums(mat_stellarscope>0)
avgExprPerExprCell <- sumExprCommonTEs/nCellsExprTEs

TEcorrsCommonTEs <- list()

# compute spearman correlation of stellarscope and soloTE counts
TEcorrsCommonTEs[["Stellarscope_SoloTE"]] <- sapply(1:nrow(mat_stellarscope), 
 ↪FUN = function(i){
  cor(mat_stellarscope[i,], mat_soloTE[i,], method="spearman")
})

# compute spearman correlation of stellarscope and STARsolo counts
TEcorrsCommonTEs[["Stellarscope_STARsolo"]] <- sapply(1:nrow(mat_stellarscope), 
 ↪FUN = function(i){
  cor(mat_stellarscope[i,], mat_STARsolo[i,], method="spearman")
})
```

```
# compute spearman correlation of stellarscope and STARsolo counts
TEcorrsCommonTEs[["STARsolo_SoloTE"]] <- sapply(1:nrow(mat_STARsolo), FUN =␣
↪function(i){
  cor(mat_STARsolo[i,], mat_soloTE[i,], method="spearman")
})
```

[ ]:
```
summary(TEcorrsCommonTEs[["Stellarscope_SoloTE"]])
summary(TEcorrsCommonTEs[["Stellarscope_STARsolo"]])
summary(TEcorrsCommonTEs[["STARsolo_SoloTE"]])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.1100  0.8065  0.9167  0.8686  1.0000  1.0000

    Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
-0.09339  0.80646  0.89500  0.86960  1.00000  1.00000

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.1140  0.6567  0.8088  0.7661  0.9194  1.0000
```

3 Checkpoint

[6]:
```
#save.image(paste0("workspaces/
↪afterCorrelations_wSTAR_thr2percCells_",dataset_id,".RData"))
load(paste0("workspaces/afterCorrelations_wSTAR_thr2percCells_",dataset_id,".
↪RData"))
rm(mat_stellarscope)
rm(mat_soloTE)
rm(mat_STARsolo)
gc()
```

A matrix: 2 × 6 of type dbl

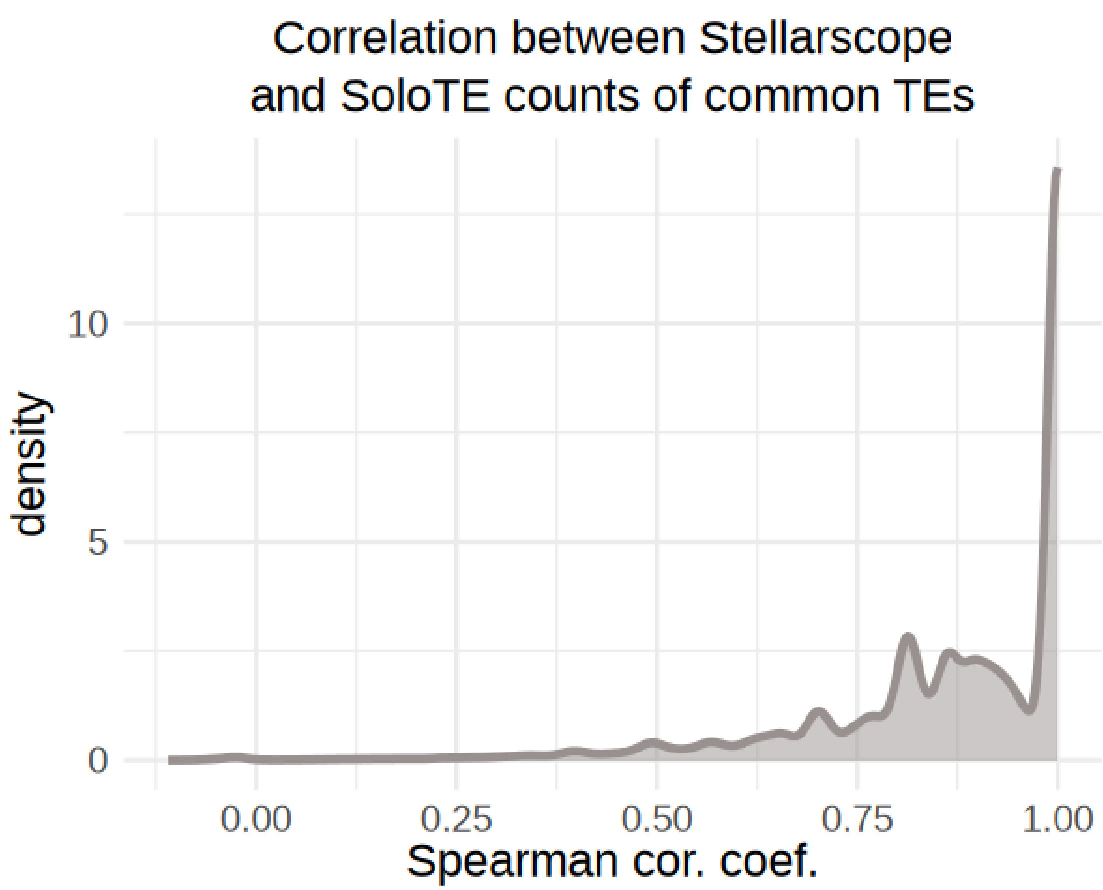
	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	27132163	1449.1	46624008	2490.0	27147683	1449.9
Vcells	231552036	1766.7	348374963	2657.9	311319318	2375.2

4 Post

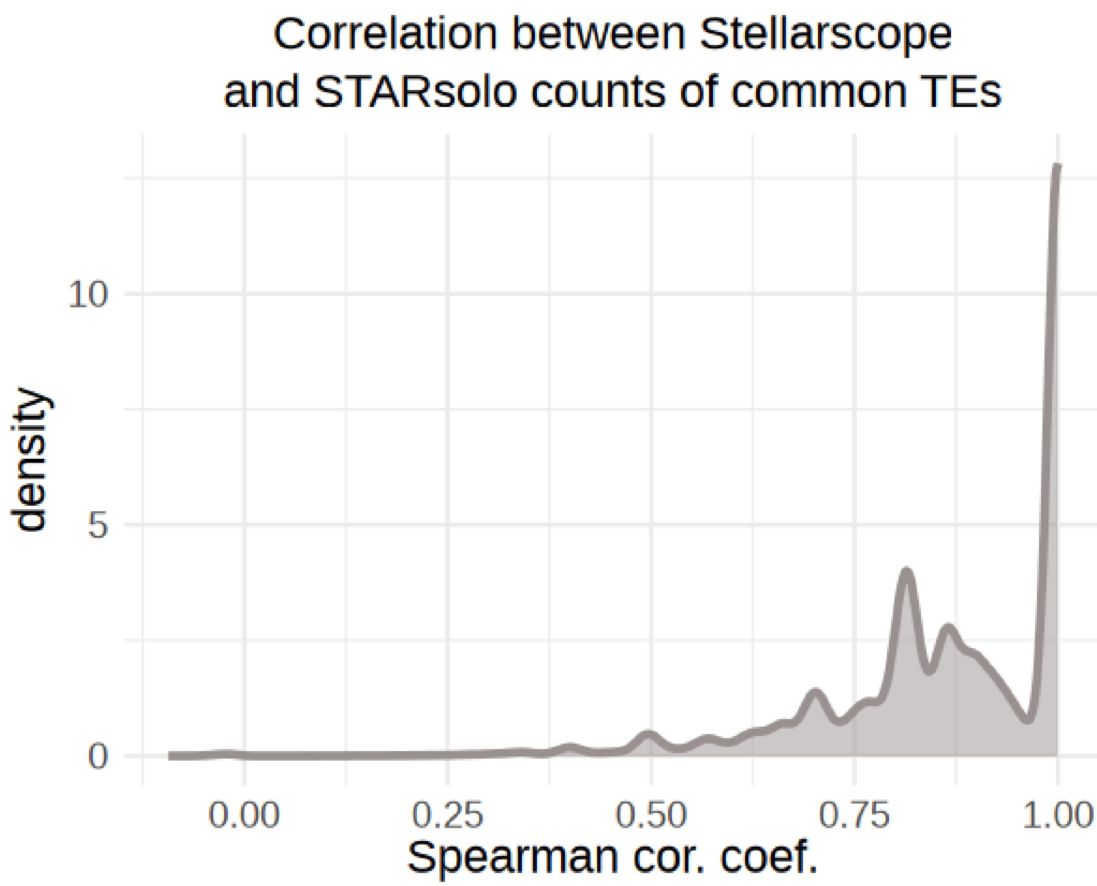
[7]:
```
options(repr.plot.width=5, repr.plot.height=4)

for(combo in names(TEcorrsCommonTEs)){

        tools <- unlist(strsplit(combo, "_"))
        show(
                ggplot() + aes(x=TEcorrsCommonTEs[[combo]]) +
                        geom_density(color="#99918F", fill=alpha("#99918F",0.
↪5), linewidth = 1) +
                        ggtitle(paste0("Correlation between ", tools[1]),
```

```
                              subtitle = paste0("and ", tools[2], " counts of␣
↪common TEs")) +
                    xlab("Spearman cor. coef.") +
                    theme_minimal() +
                    theme(text=element_text(size=14.5),
                              plot.title = element_text(size=14.5, hjust=0.
↪5),
                              plot.subtitle = element_text(size=14.5, hjust=0.
↪5))
        )
                ggsave(paste0("figures_",dataset_id,"/
↪spearmanCorrelation_density_commonTEs_",combo,".pdf"), device = "pdf")
}
```

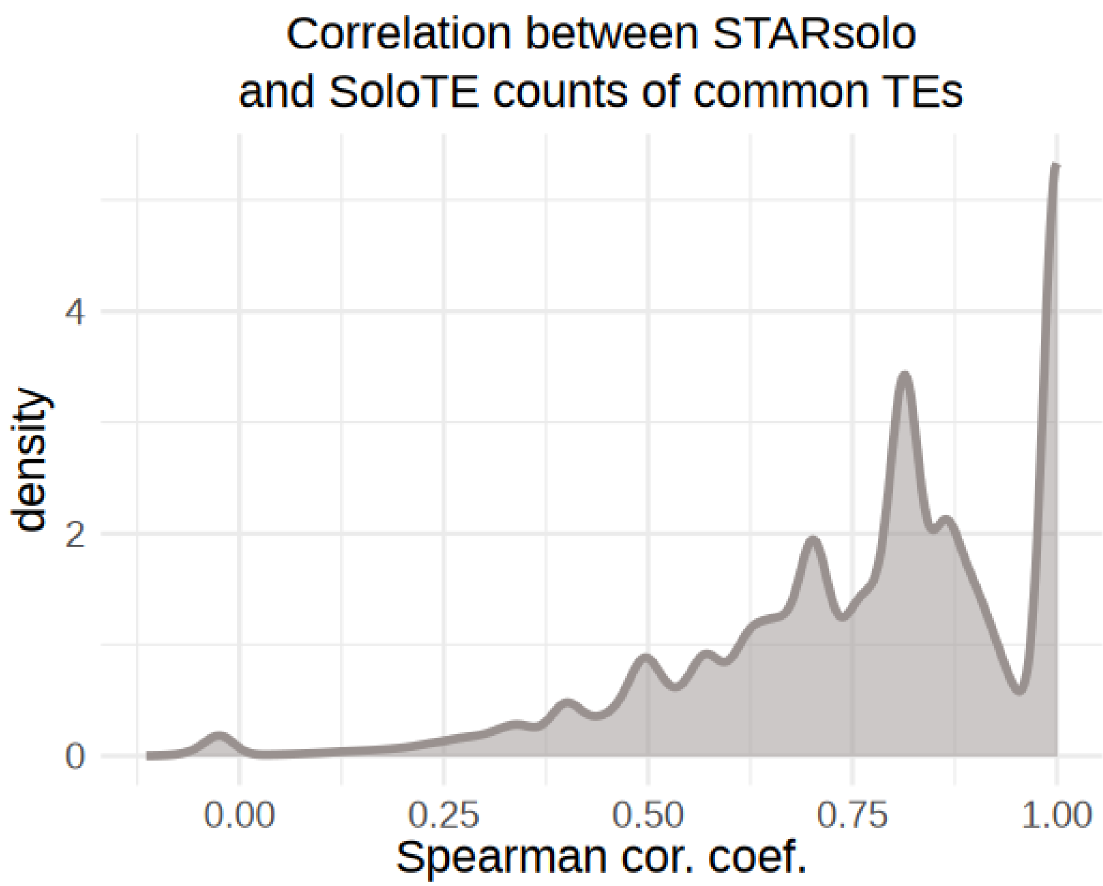
```
Saving 6.67 x 6.67 in image
```




```
Saving 6.67 x 6.67 in image
```




```
Saving 6.67 x 6.67 in image
```



[8]:
```r
# define age breaks for bins
breaks <- c(0, 2, 5, 10, 15, 25, 40, 60, Inf)

summary(conversionTable$mya)

conversionTable$age_bin <- cut(
  conversionTable$mya,
  breaks = breaks,
  labels = c("0-2", "2-5", "5-10", "10-15", "15-25", "25-40", "40-60", "60+"),
  right = FALSE,
  include.lowest = TRUE
)

table(conversionTable$age_bin)

table(conversionTable$age_bin[is.na(conversionTable$mya)])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
   0.00   16.67   27.53   28.36   39.12  142.90  311572 
```

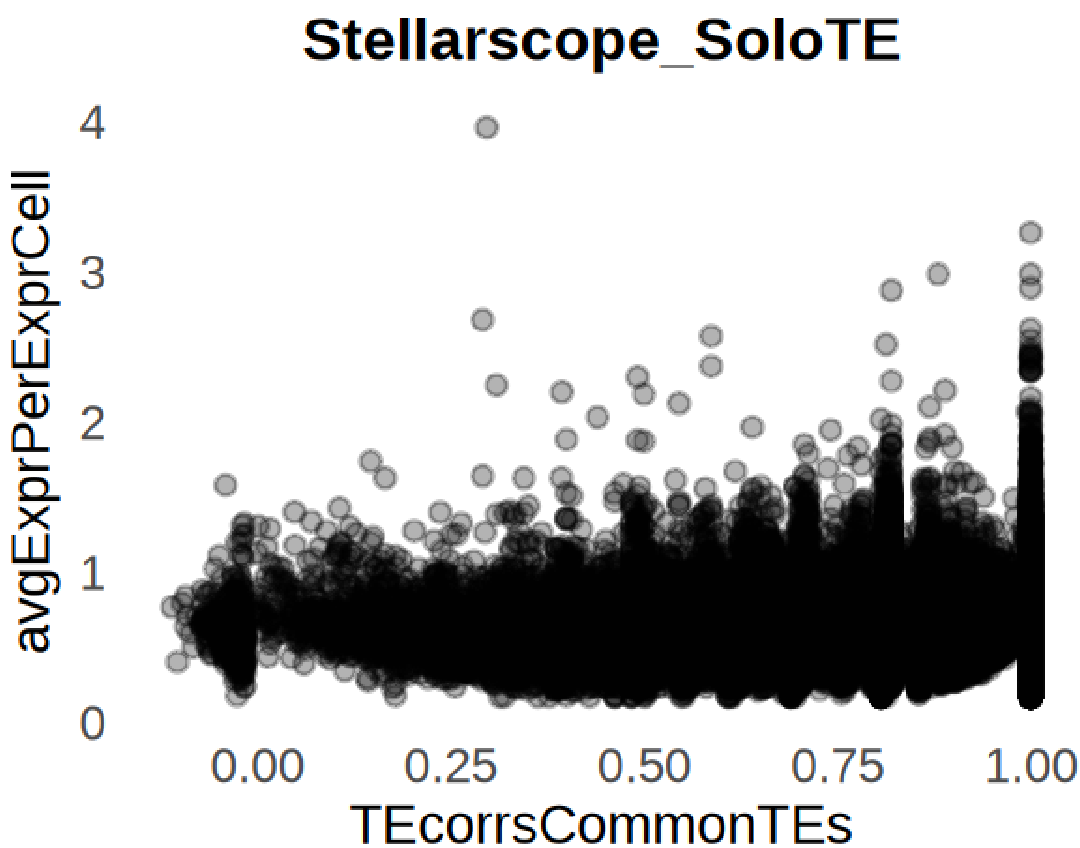
```
    0-2      2-5     5-10    10-15    15-25    25-40    40-60      60+
  13303    39956   248223   632189  1102057  1438502  1028605    26428
```

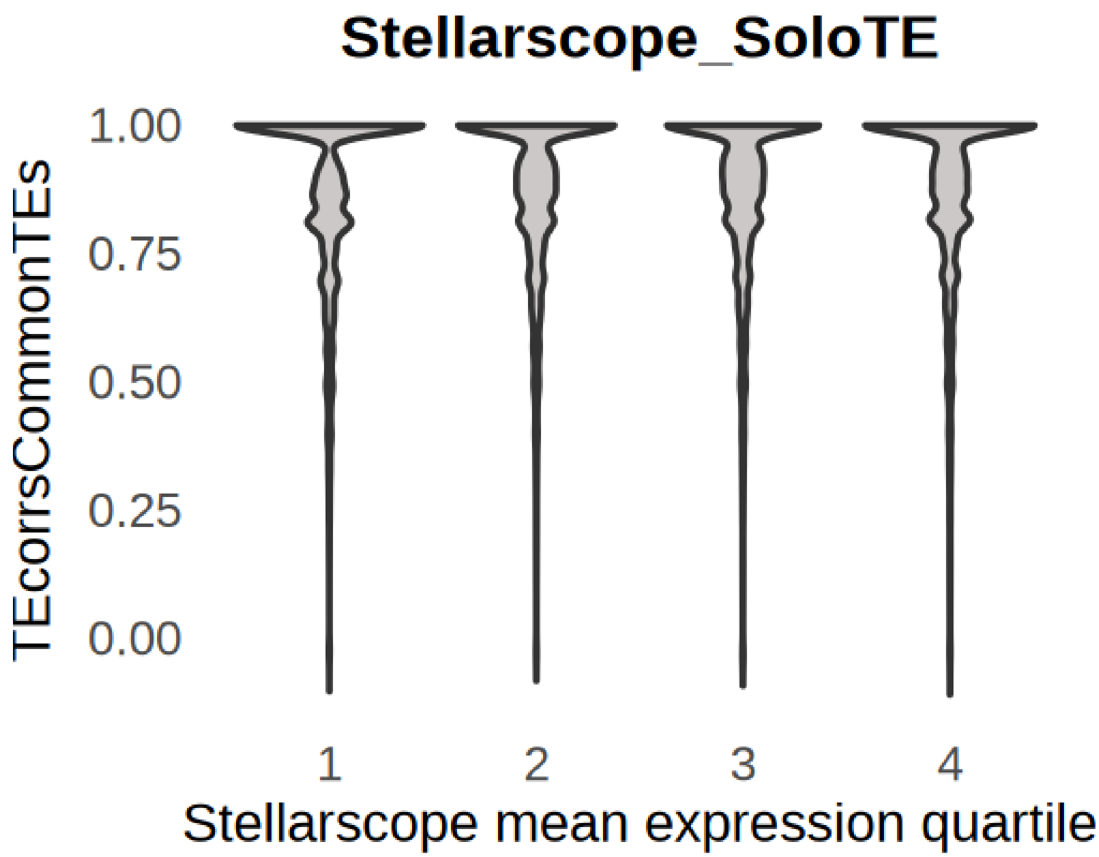
```
  0-2   2-5  5-10 10-15 15-25 25-40 40-60   60+
    0     0     0     0     0     0     0     0
```

[18]:
```
options(repr.plot.width=5, repr.plot.height=4)

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))

  df <- cbind(avgExprPerExprCell, TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("avgExprPerExprCell", "TEcorrsCommonTEs")

  cat("Correlation between correlations and avg expression (", combo,") : ",
    cor(avgExprPerExprCell, TEcorrsCommonTEs[[combo]], method="spearman"),"\n")


  show(ggplot(df, aes(x=TEcorrsCommonTEs, y=avgExprPerExprCell)) +␣
↪ggtitle(combo) +
        geom_point(alpha= 0.3))

  # divide by expr quartiles
  df$quartile <- "4"
  df$quartile[df$avgExprPerExprCell < quantile(df$avgExprPerExprCell, 0.75, na.
↪rm=TRUE)] <- "3"
  df$quartile[df$avgExprPerExprCell < quantile(df$avgExprPerExprCell, 0.5, na.
↪rm=TRUE)] <- "2"
  df$quartile[df$avgExprPerExprCell < quantile(df$avgExprPerExprCell, 0.25, na.
↪rm=TRUE)] <- "1"

  show(ggplot(df, aes(x=quartile, y=TEcorrsCommonTEs)) + ggtitle(combo) +
    geom_violin(fill=alpha("#99918F",0.5)) +
    theme(text=element_text(size=15)) +
    xlab("Stellarscope mean expression quartile") )

  # divide by correlation quartiles
  df$quartile <- "4"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
↪rm=TRUE)] <- "3"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
↪rm=TRUE)] <- "2"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
↪rm=TRUE)] <- "1"
```

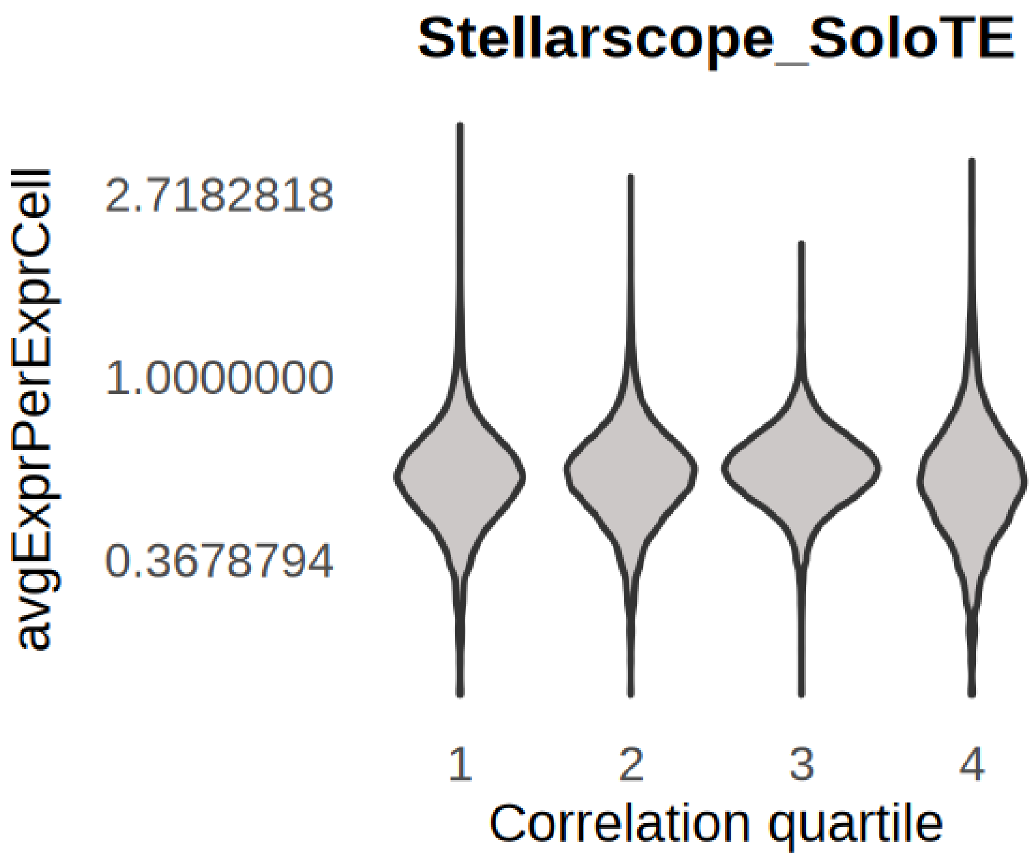
```
  show(ggplot(df, aes(x=quartile, y=avgExprPerExprCell)) +
    geom_violin(fill=alpha("#99918F",0.5)) + ggtitle(combo) +
    theme(text=element_text(size=15)) +
    scale_y_continuous(transform = "log") +
    xlab("Correlation quartile") )
}
```

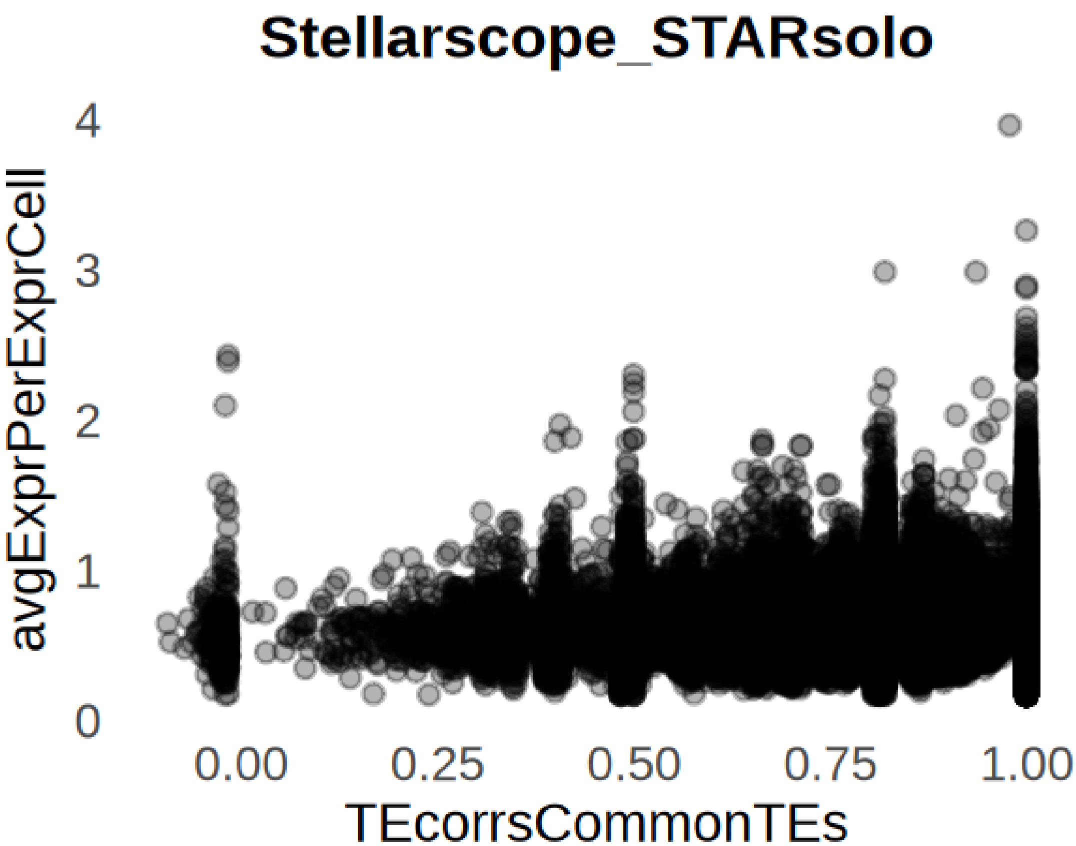
```
Correlation between correlations and avg expression ( Stellarscope_SoloTE ) :
-0.01250458
```

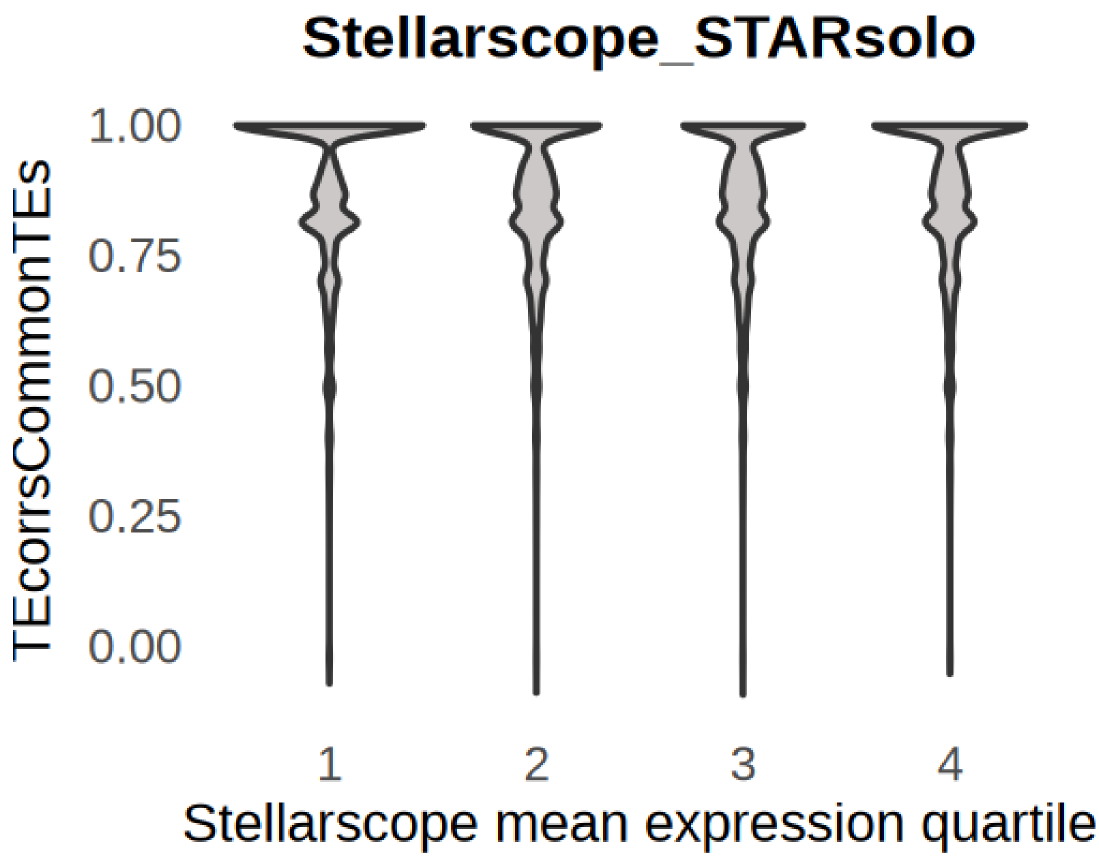




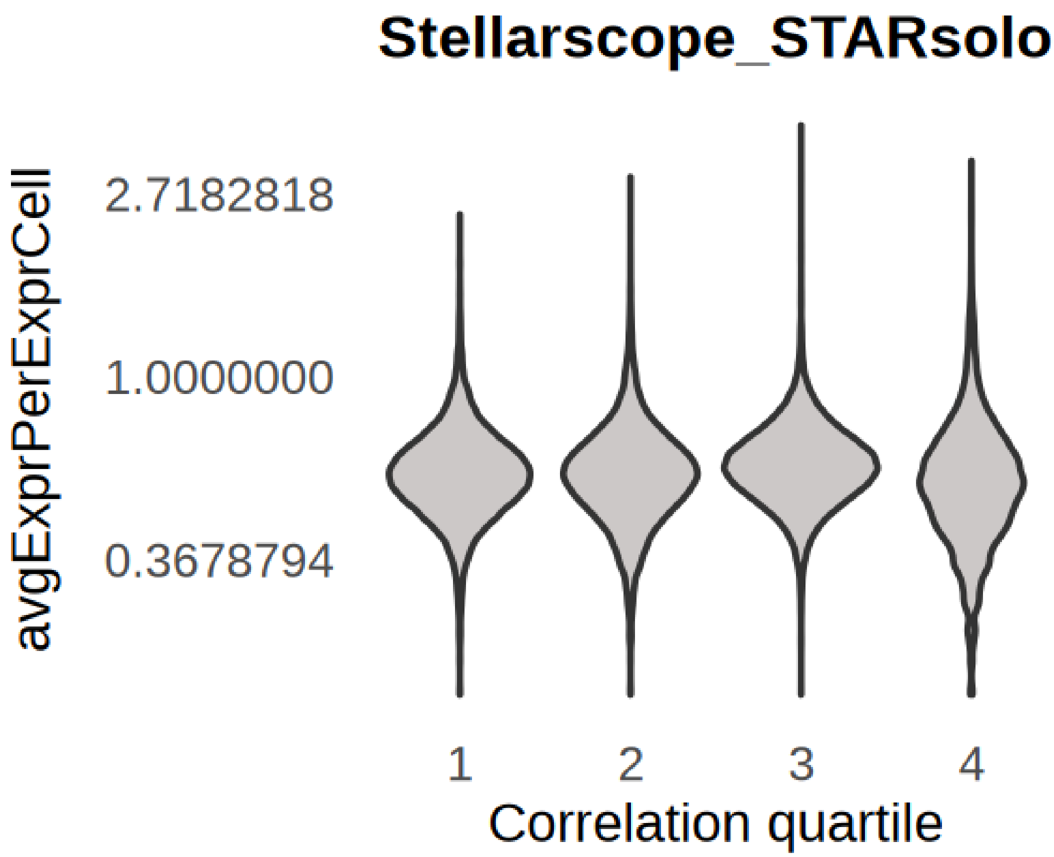

```
Correlation between correlations and avg expression ( Stellarscope_STARsolo ) :
-0.04608201
```

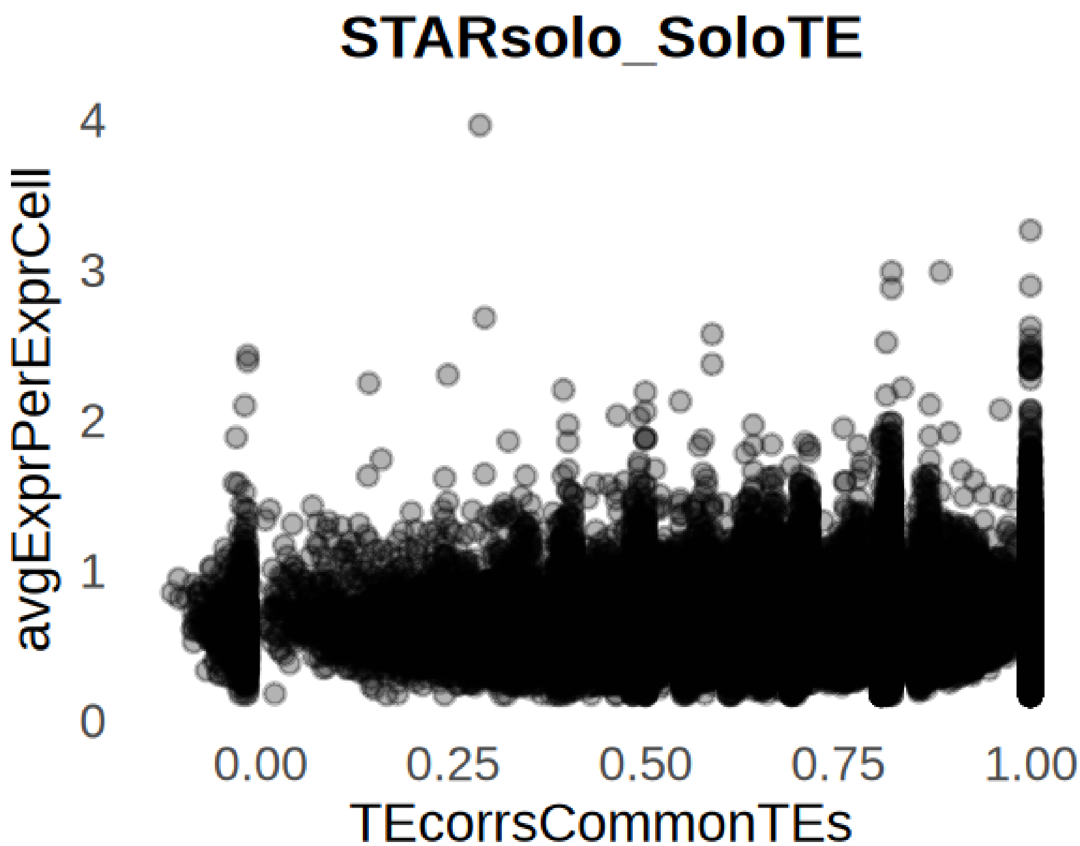

Stellarscope_SoloTE
avgExprPerExprCell
2.7182818
1.0000000
0.3678794
1
2
3
4
Correlation quartile


Stellarscope_STARsolo
avgExprPerExprCell
4
3
2
1
0
0.00
0.25
0.50
0.75
1.00
TEcorrsCommonTEs

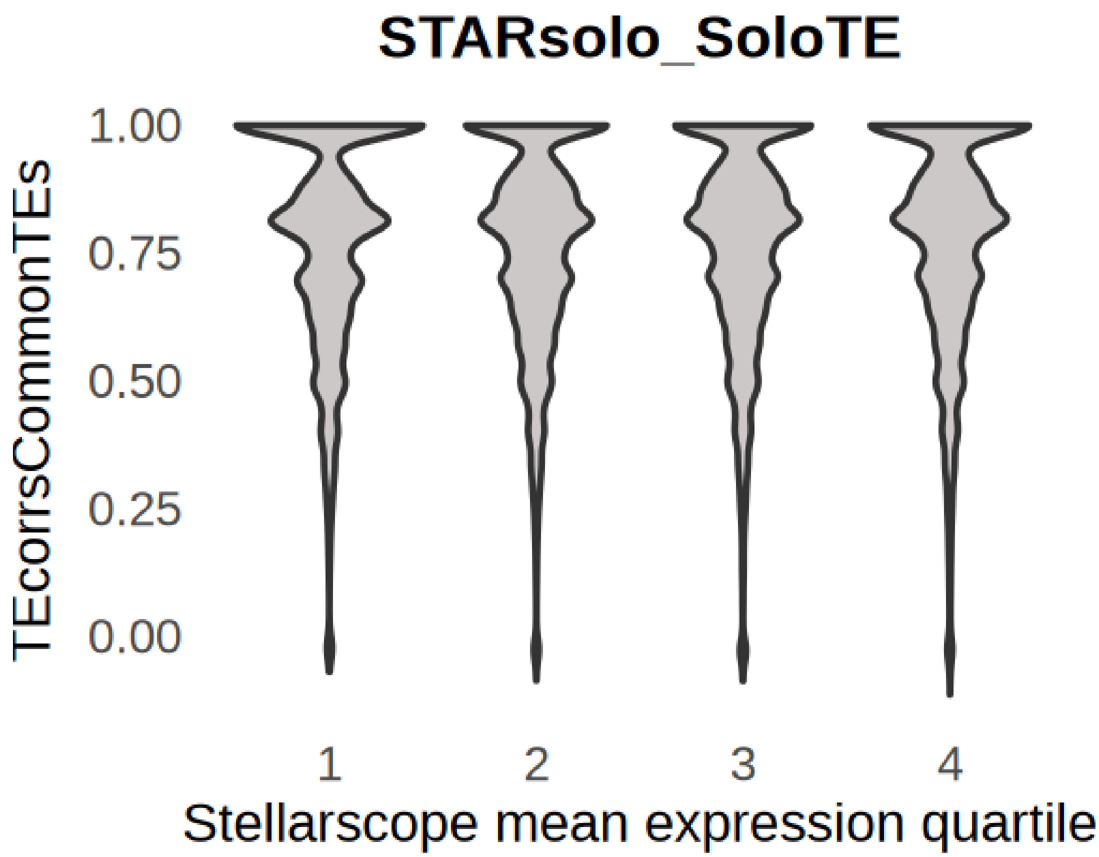


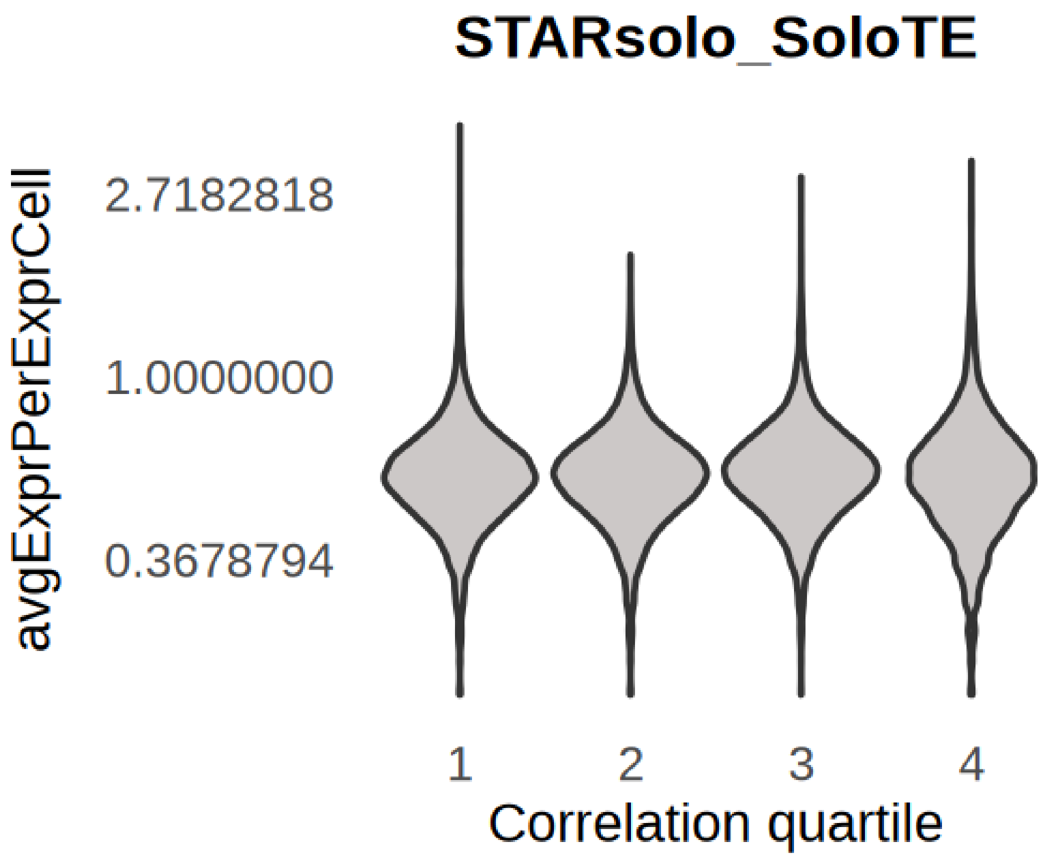

```
Correlation between correlations and avg expression ( STARsolo_SoloTE ) :
-0.002652486
```


Stellarscope_STARsolo
avgExprPerExprCell
2.7182818
1.0000000
0.3678794
1
2
3
4
Correlation quartile



STARsolo_SoloTE
avgExprPerExprCell
4
3
2
1
0
0.00
0.25
0.50
0.75
1.00
TEcorrsCommonTEs


STARsolo_SoloTE
1.00
0.75
0.50
0.25
0.00
TEcorrsCommonTEs
1
2
3
4
Stellarscope mean expression quartile



4.1 Correlation by age

[48]:
```
library(ragg)
options(device = "ragg::agg_png")
```

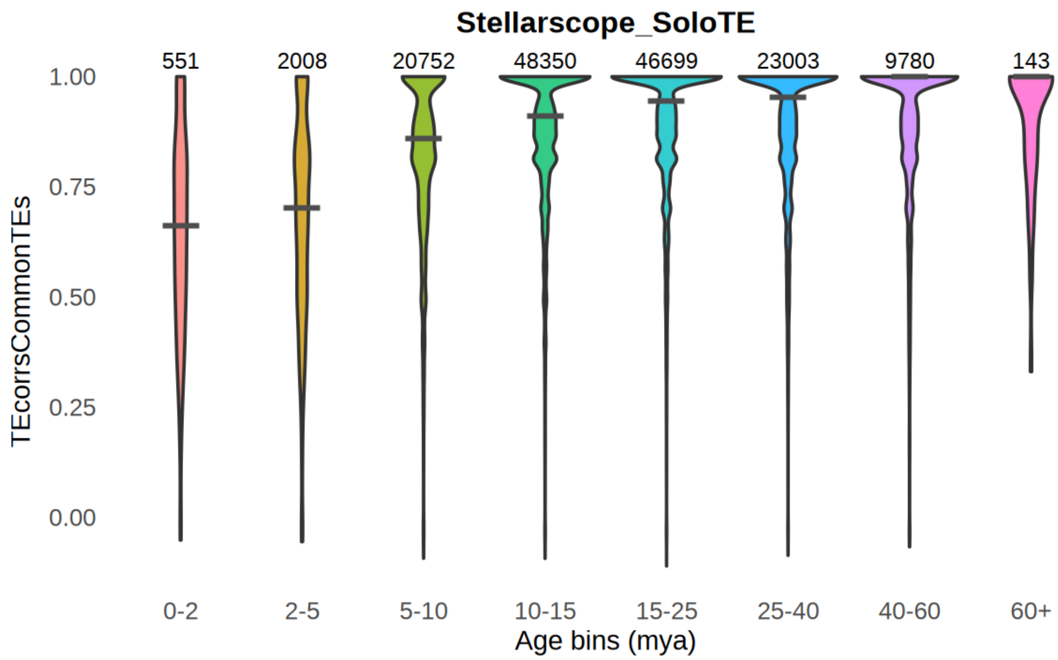
```
Warning message:
"package ‘ragg’ was built under R version 4.4.3"
```

[59]:
```
options(repr.plot.width=10, repr.plot.height=6)
library(ggbeeswarm)

library(ragg)
options(device = "ragg::agg_png")

dfsCorrelationAge <- list()

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))
```

```
  df <- cbind(rownames(matSubs[[tools[1]]]), TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("V1", "TEcorrsCommonTEs")

  df$ageClass <- conversionTable$ageClass[match(df$V1, conversionTable$stellarscopeID)]
  df$age_bin <- conversionTable$age_bin[match(df$V1, conversionTable$stellarscopeID)]
  df$mya <- conversionTable$mya[match(df$V1, conversionTable$stellarscopeID)]
  df$TEcorrsCommonTEs <- as.numeric(df$TEcorrsCommonTEs)
  cor(df$mya, df$TEcorrsCommonTEs, method="spearman")


  show(  ggplot(df, aes(x=age_bin, y=TEcorrsCommonTEs, fill=age_bin)) +
        geom_violin(alpha = 0.8) +
            stat_summary(fun.data = function(x) data.frame(y = max(x), label = length(x)),
             geom = "text", vjust = -0.5, size = 5, color = "black") +
             stat_summary(
            fun = median,
            geom = "crossbar",
            width = 0.3,
            color = "grey30",
            linewidth = 0.5
      ) +
        theme_paper() +
        #scale_fill_manual(values=colorAge) +
        guides(fill = "none") +
        theme(text=element_text(size=18)) +
        xlab("Age bins (mya)") +
        ggtitle(combo)
  )
ggsave(paste0("figures_",dataset_id,"/correlation_byAge_Bin_",combo,"_violin.pdf"), width=8, height=6)

# if(combo=="Stellarscope_SoloTE"){
# show(  ggplot(df, aes(x=age_bin, y=TEcorrsCommonTEs, color=age_bin)) +
#         geom_beeswarm(alpha = 0.7,cex = 0.2, size=0.5) +
#         #geom_boxplot(outlier.shape = NA, width = 0.3, color = "gray40", fill=NA, alpha = 0.3) +
#       stat_summary(
#             fun = median,
#             geom = "crossbar",
#             width = 0.3,
#             color = "grey30",
#             linewidth = 0.5
#       ) +
```

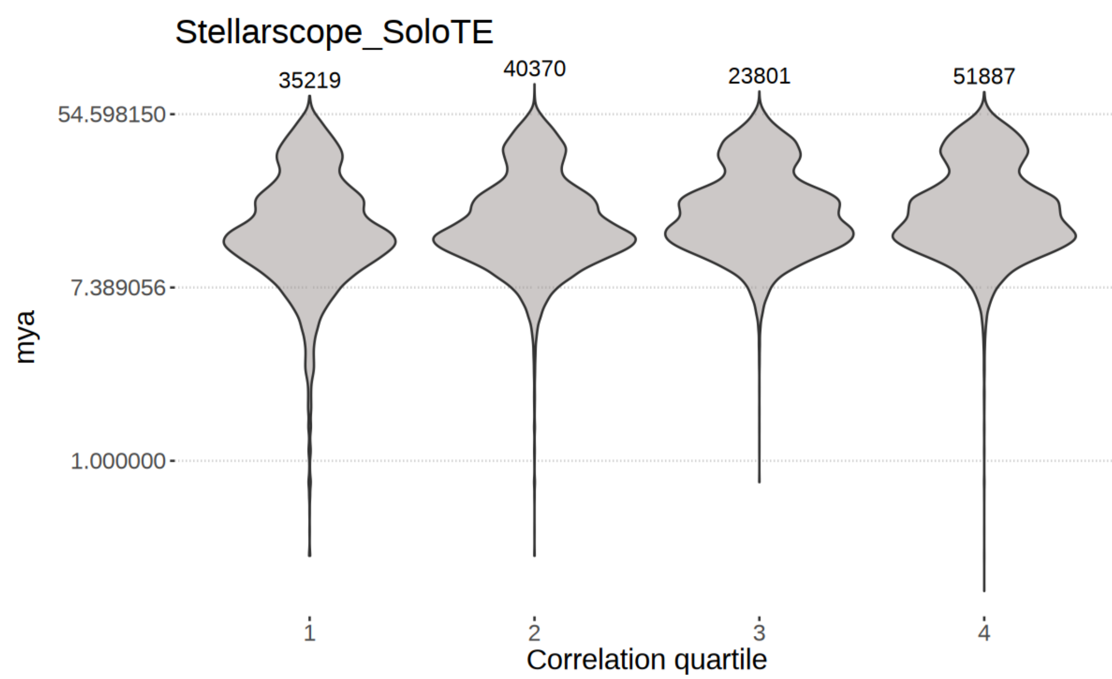
```
#             stat_summary(fun.data = function(x) data.frame(y = max(x), label␣
↪= length(x)),
#                   geom = "text", vjust = -0.5, size = 5, color = "black") +
#          theme_paper() +
#          #scale_fill_manual(values=colorAge) +
#          guides(color = "none") +
#          theme(text=element_text(size=18)) +
#          xlab("Age bins (mya)") +
#          ylab("Spearman cor. coef.") +
#          ggtitle(combo)
#   )
# }
 ggsave(paste0("figures_",dataset_id,"/correlation_byAge_Bin_",combo,".pdf"),␣
↪width=8, height=6)

 show(  ggplot(df, aes(x=ageClass, y=TEcorrsCommonTEs, fill=ageClass)) +
       geom_violin(alpha = 0.5) +
           stat_summary(fun.data = function(x) data.frame(y = max(x), label =␣
↪length(x)),
            geom = "text", vjust = -0.5, size = 5, color = "black") +
       theme_pubclean() +
       scale_fill_manual(values=colorAge) +
       theme(text=element_text(size=18)) +
       xlab("Age class") +
       ggtitle(combo)
 )

 # divide by correlation quartiles
 df$quartile <- "4"
 df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
↪rm=TRUE)] <- "3"
 df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
↪rm=TRUE)] <- "2"
 df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
↪rm=TRUE)] <- "1"

 show(ggplot(df, aes(x=quartile, y=mya)) +
       geom_violin(fill=alpha("#99918F",0.5)) +
           stat_summary(fun.data = function(x) data.frame(y = max(x), label =␣
↪length(x)),
            geom = "text", vjust = -0.5, size = 5) +
       theme_pubclean() +
       theme(text=element_text(size=18)) +
       scale_y_continuous(transform = "log") +
       xlab("Correlation quartile") +
       ggtitle(combo)
```

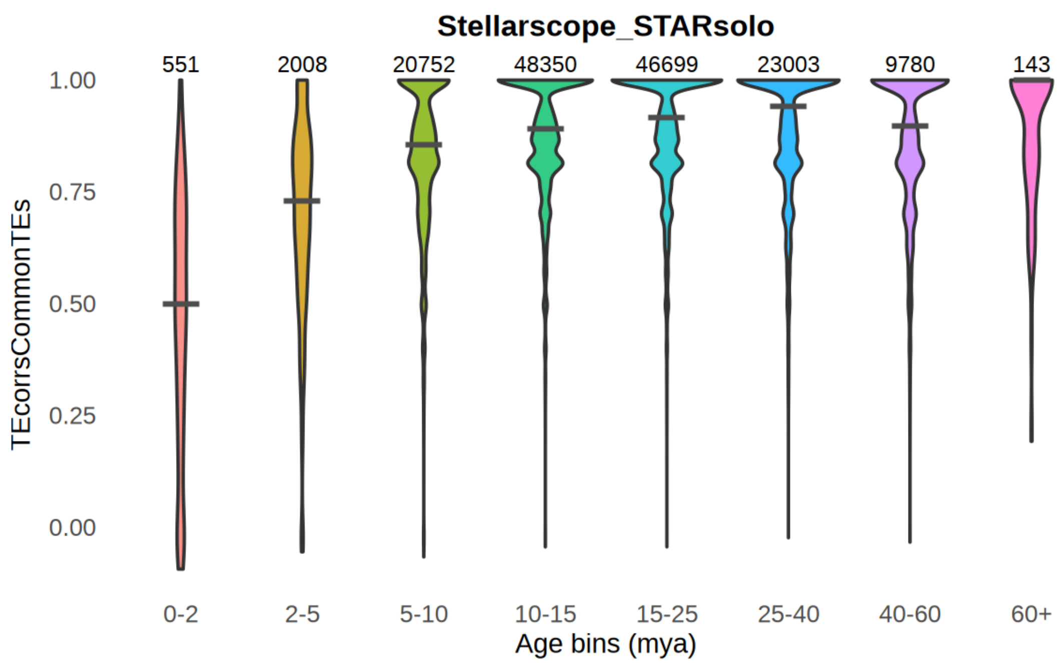
```
  )

  dfsCorrelationAge[[combo]] <- df
}
```



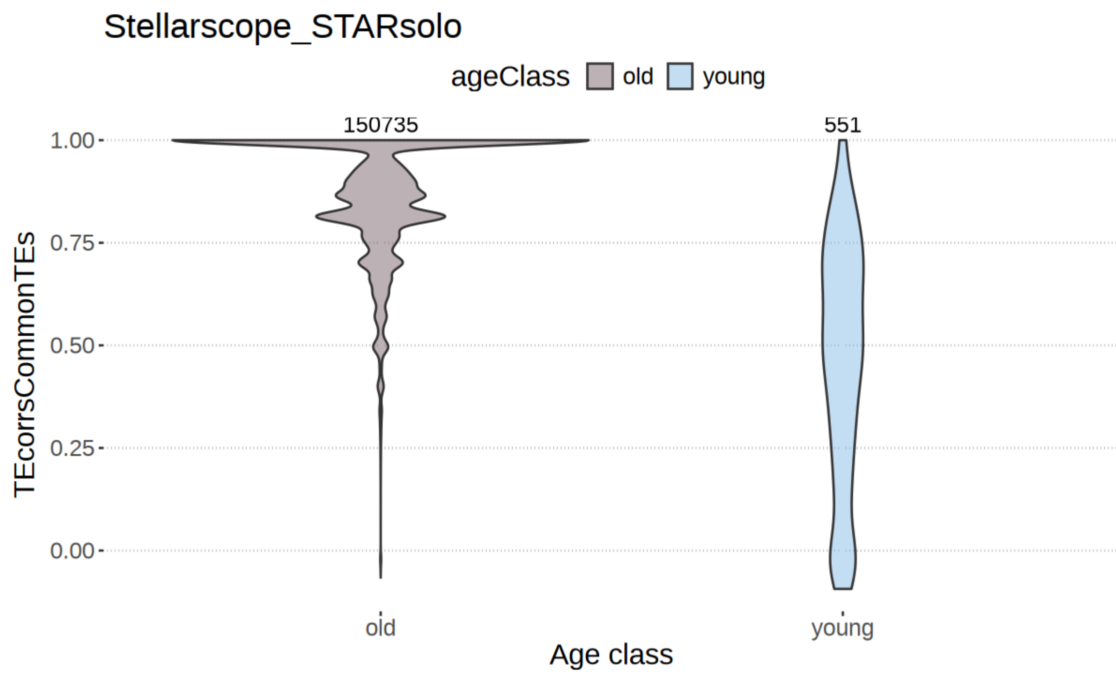

```
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_ydensity()`)."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_summary()`)."
```

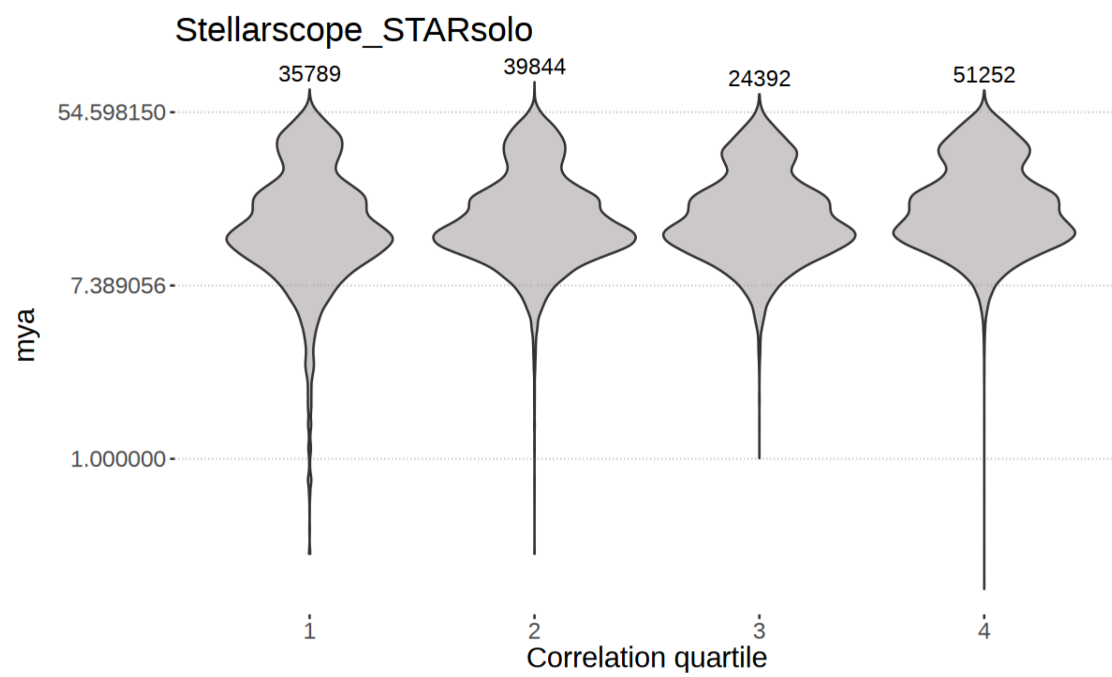

Stellarscope_SoloTE
ageClass
old
young
150735
551
1.00
0.75
0.50
0.25
0.00
TEcorrsCommonTEs
old
young
Age class

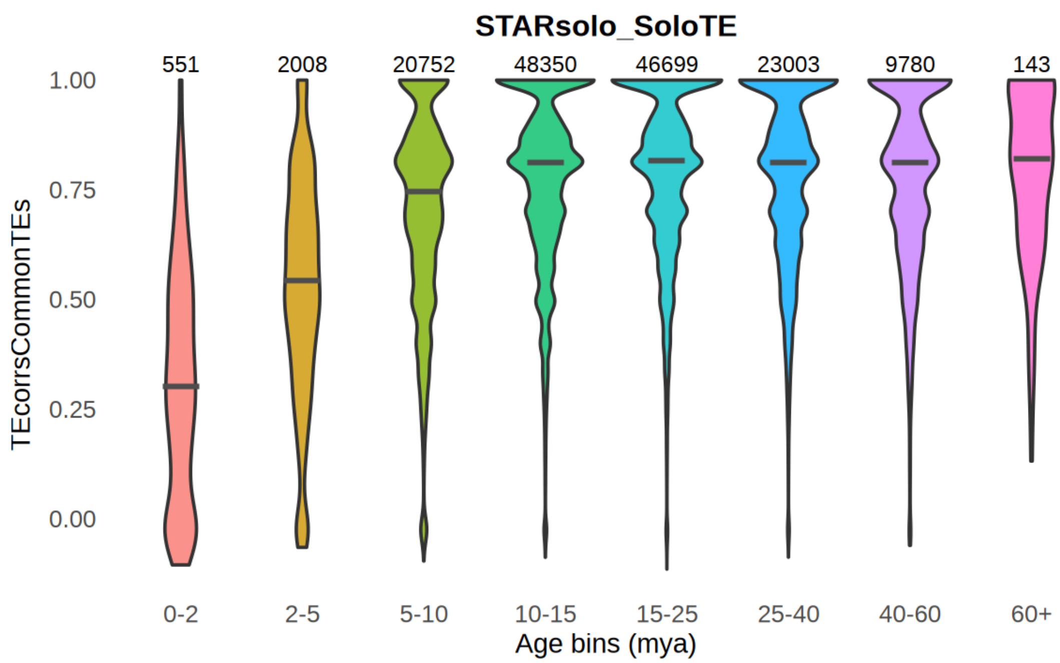

Stellarscope_SoloTE
35219
40370
23801
51887
54.598150
7.389056
1.000000
mya
1
2
3
4
Correlation quartile




```
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_ydensity()`)."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_summary()`)."
```


Stellarscope_STARsolo
ageClass old young
150735
551
1.00
0.75
0.50
0.25
0.00
TEcorrsCommonTEs
old
young
Age class

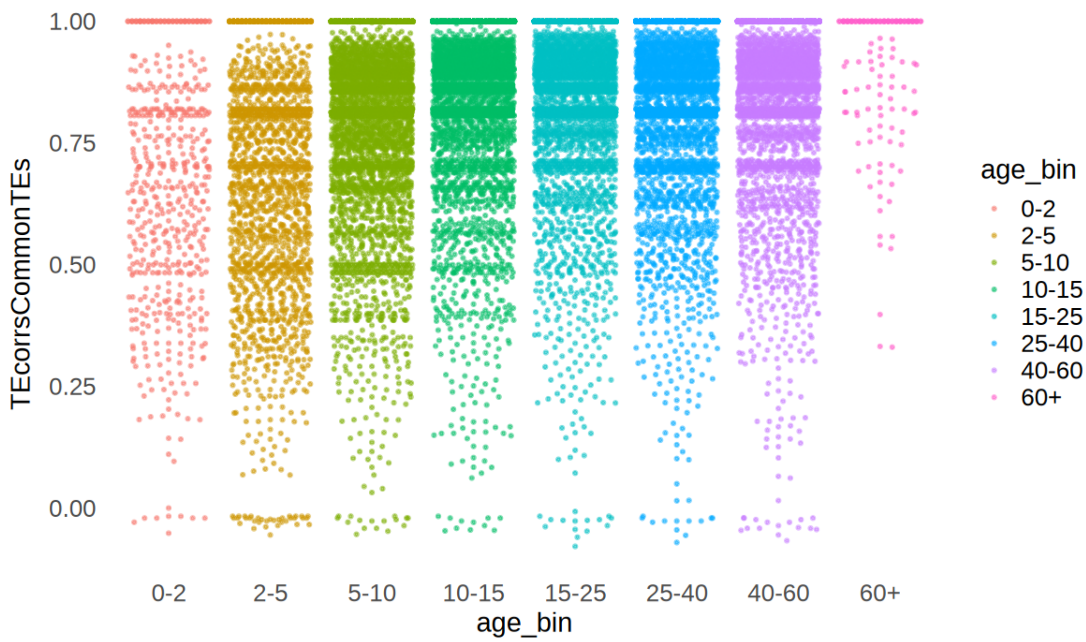

Stellarscope_STARsolo
35789
39844
24392
51252
54.598150
7.389056
1.000000
mya
1
2
3
4
Correlation quartile




```
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_ydensity()`)."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_summary()`)."
```

[57]:
```
library(dplyr)
set.seed(42)
df_plot <- df %>% group_by(age_bin) %>% slice_sample(n = 5000) %>% ungroup()
```

```
ggplot(df_plot, aes(x=age_bin, y=TEcorrsCommonTEs, color=age_bin)) +
  geom_beeswarm(alpha = 0.5, cex = 1.5, size = 0.3, priority = "density",␣
 ↪corral = "wrap", corral.width = 0.8)
```



[ ]:
```
gc()
```

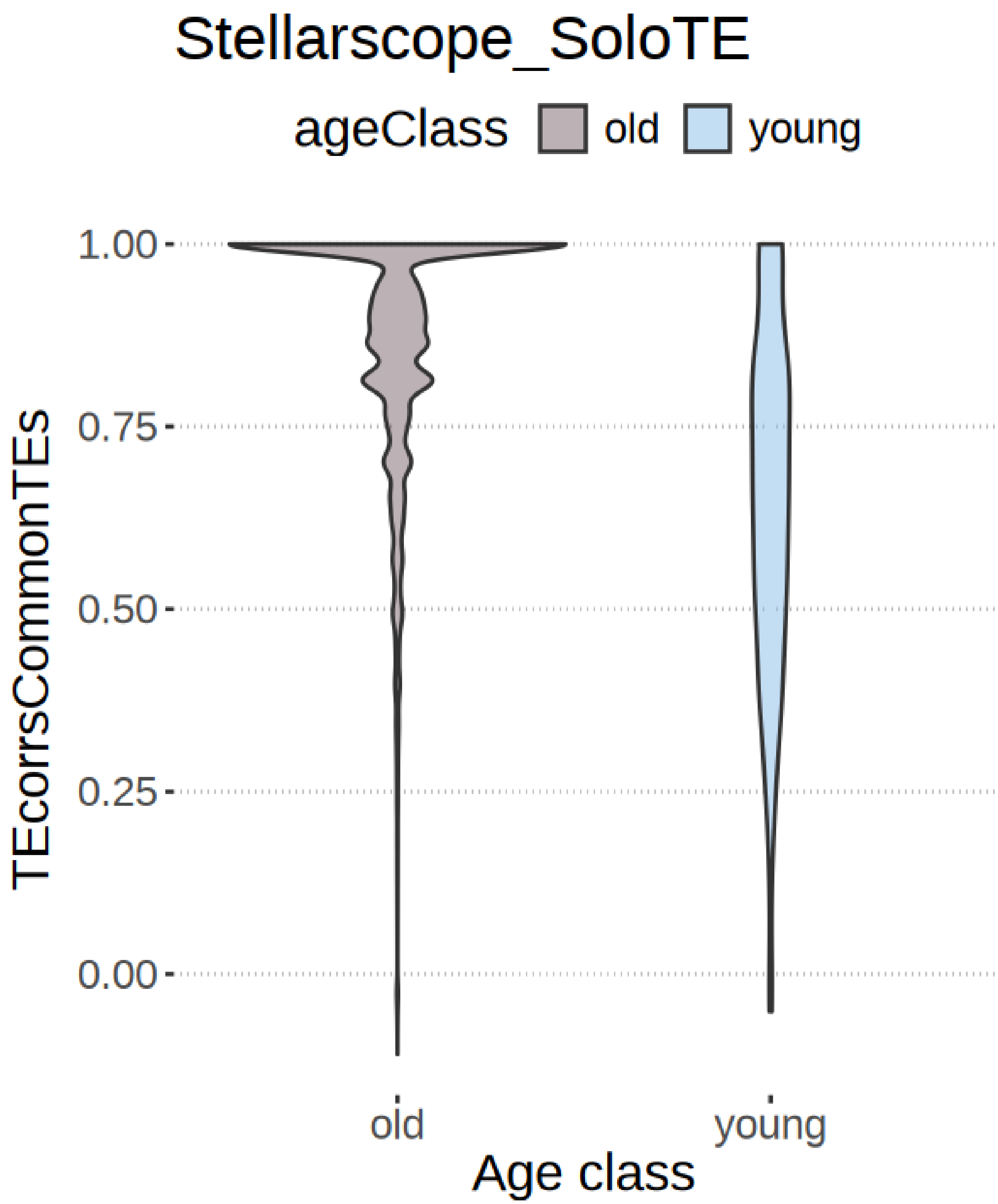
A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	27385193	1462.6	46624008	2490.0	35894211	1917.0
Vcells	236481965	1804.3	348374963	2657.9	347092909	2648.2

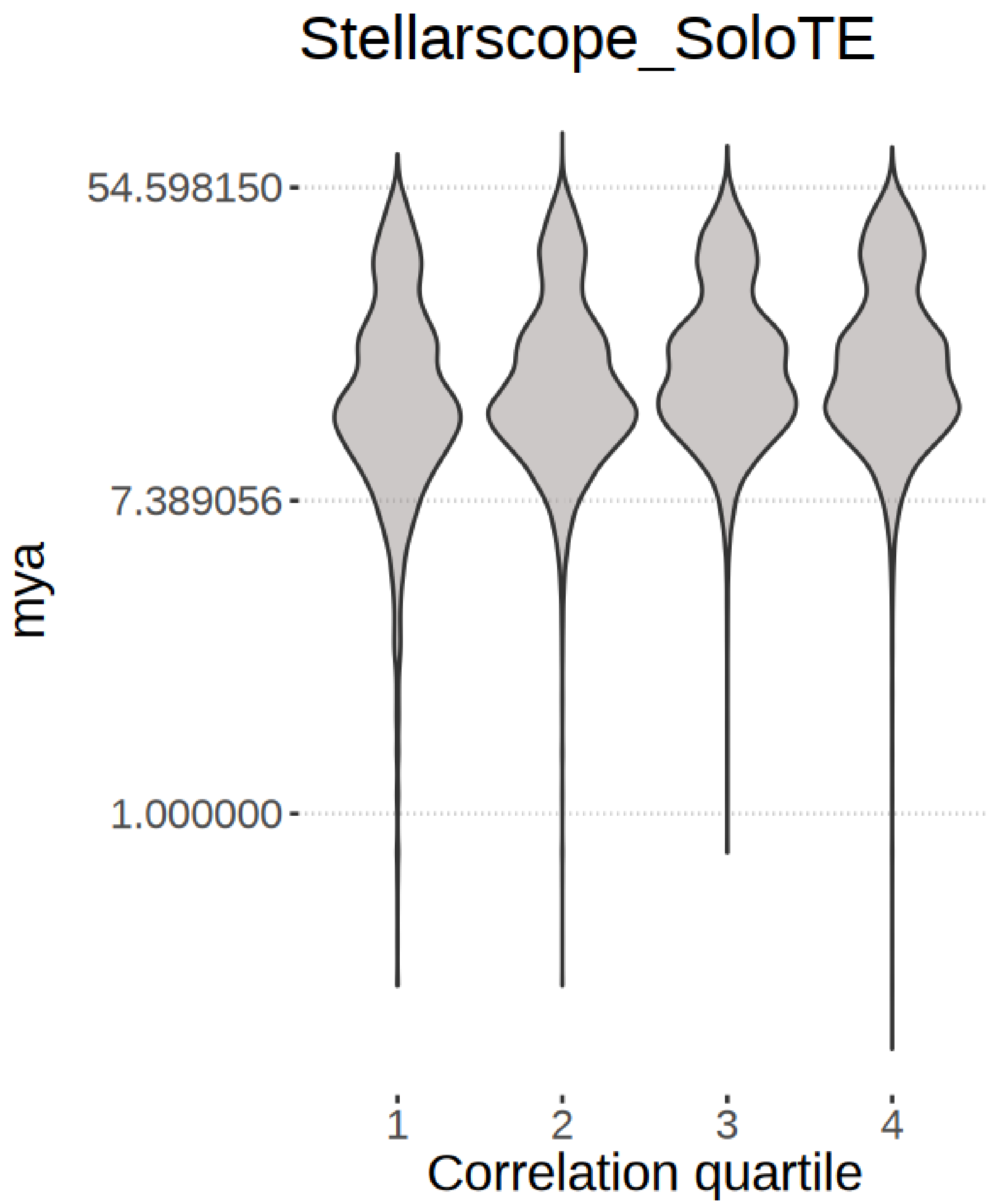
[11]:
```
options(repr.plot.width=5, repr.plot.height=6)

dfsCorrelationAge <- list()

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))

  df <- cbind(rownames(matSubs[[tools[1]]]), TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("V1", "TEcorrsCommonTEs")

  df$ageClass <- conversionTable$ageClass[match(df$V1,␣
 ↪conversionTable$stellarscopeID)]
  df$mya <- conversionTable$mya[match(df$V1, conversionTable$stellarscopeID)]
  df$TEcorrsCommonTEs <- as.numeric(df$TEcorrsCommonTEs)
```

```
  cor(df$mya, df$TEcorrsCommonTEs, method="spearman")

  options(repr.plot.width=5, repr.plot.height=6)

  show(  ggplot(df, aes(x=ageClass, y=TEcorrsCommonTEs, fill=ageClass)) +
        geom_violin(jitter=TRUE, alpha = 0.5) +
        theme_pubclean() +
        scale_fill_manual(values=colorAge) +
        theme(text=element_text(size=18)) +
        xlab("Age class") +
        ggtitle(combo)
  )


  # divide by correlation quartiles
  df$quartile <- "4"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
↪rm=TRUE)] <- "3"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
↪rm=TRUE)] <- "2"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
↪rm=TRUE)] <- "1"

  show(ggplot(df, aes(x=quartile, y=mya)) +
        geom_violin(fill=alpha("#99918F",0.5)) +
        theme_pubclean() +
        theme(text=element_text(size=18)) +
        scale_y_continuous(transform = "log") +
        xlab("Correlation quartile") +
        ggtitle(combo)
  )

  dfsCorrelationAge[[combo]] <- df
}
```

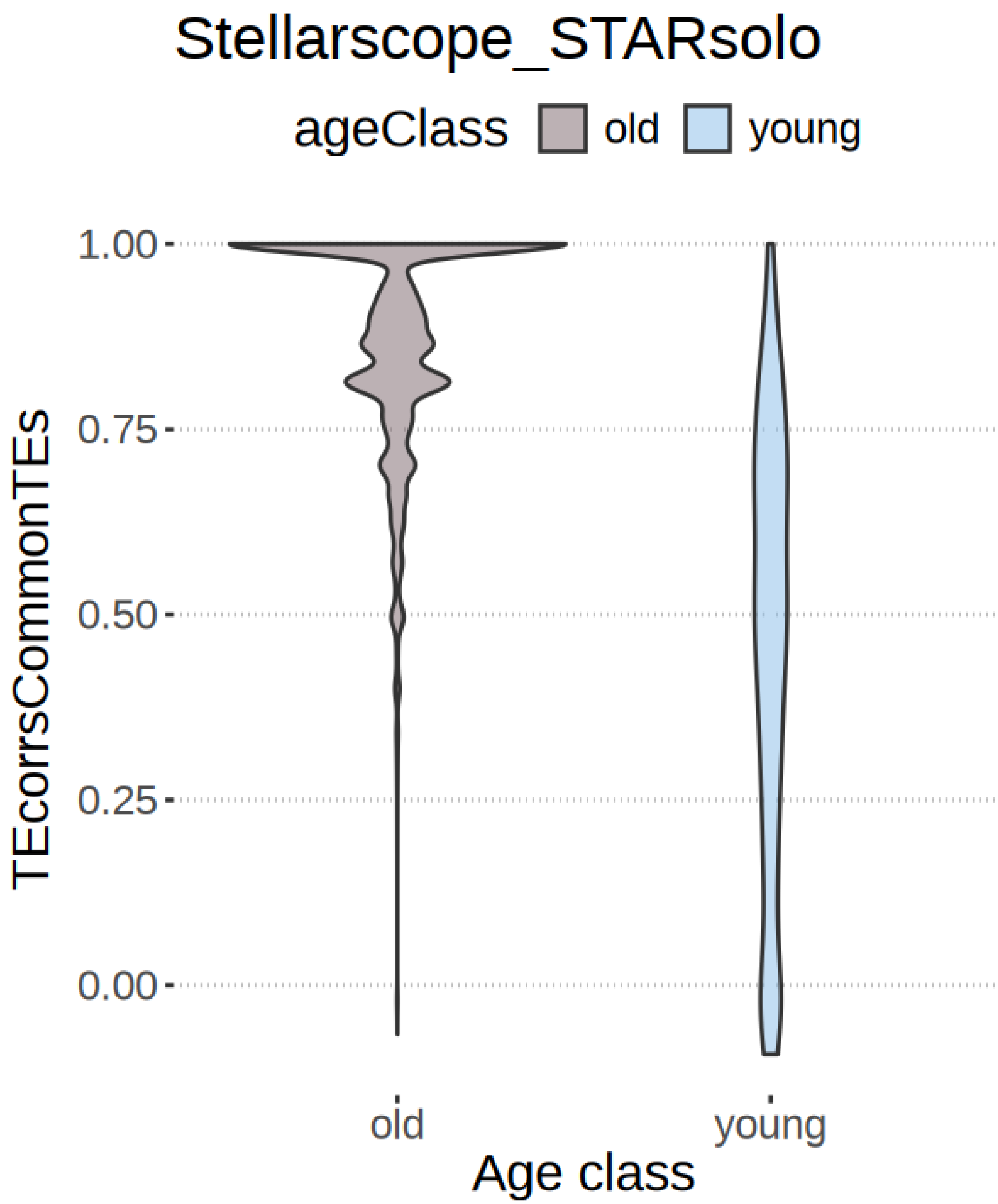
```
Warning message in geom_violin(jitter = TRUE, alpha = 0.5):
"Ignoring unknown parameters: `jitter`"
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_ydensity()`)."
```



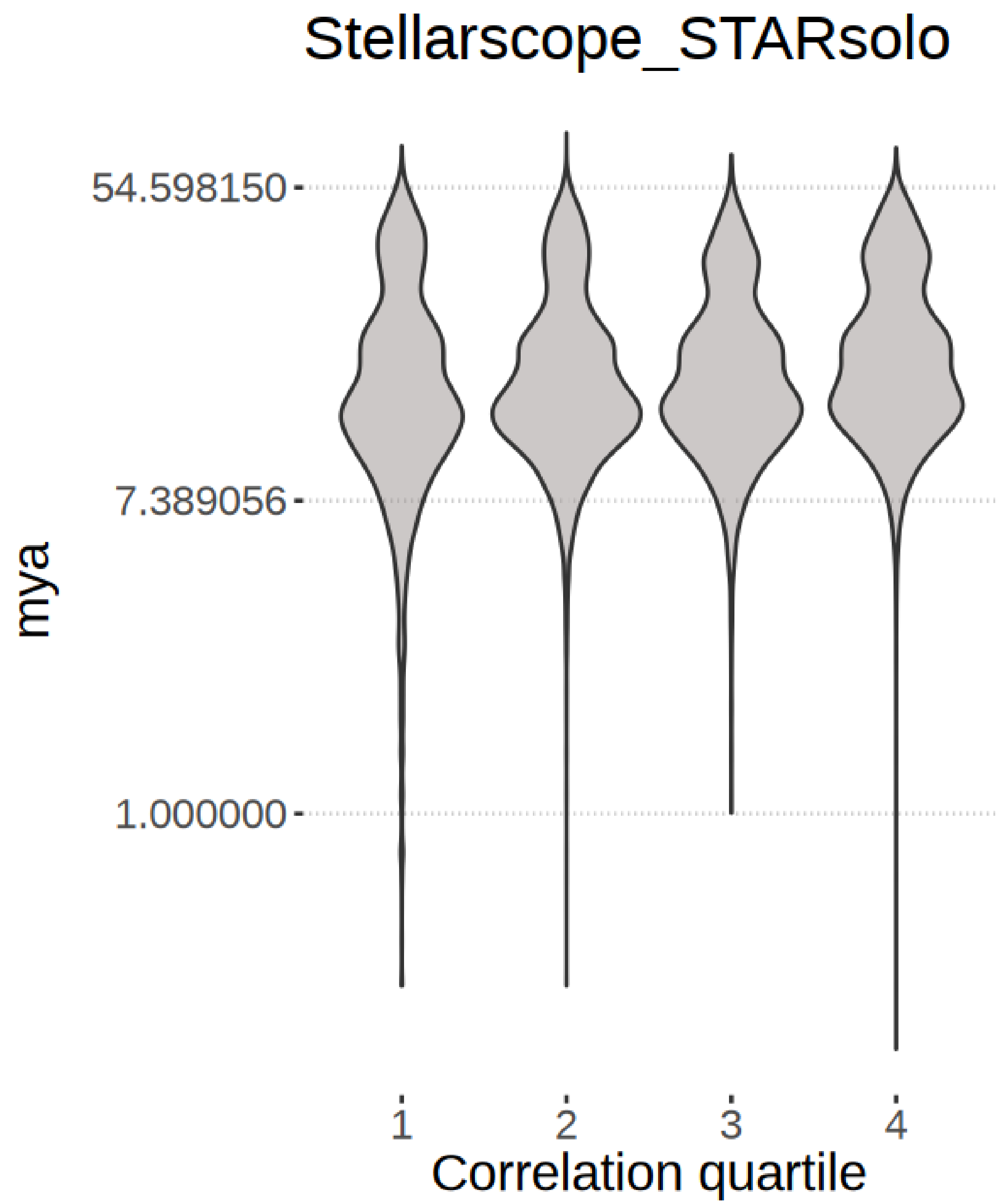

```
Warning message in geom_violin(jitter = TRUE, alpha = 0.5):
"Ignoring unknown parameters: `jitter`"
```



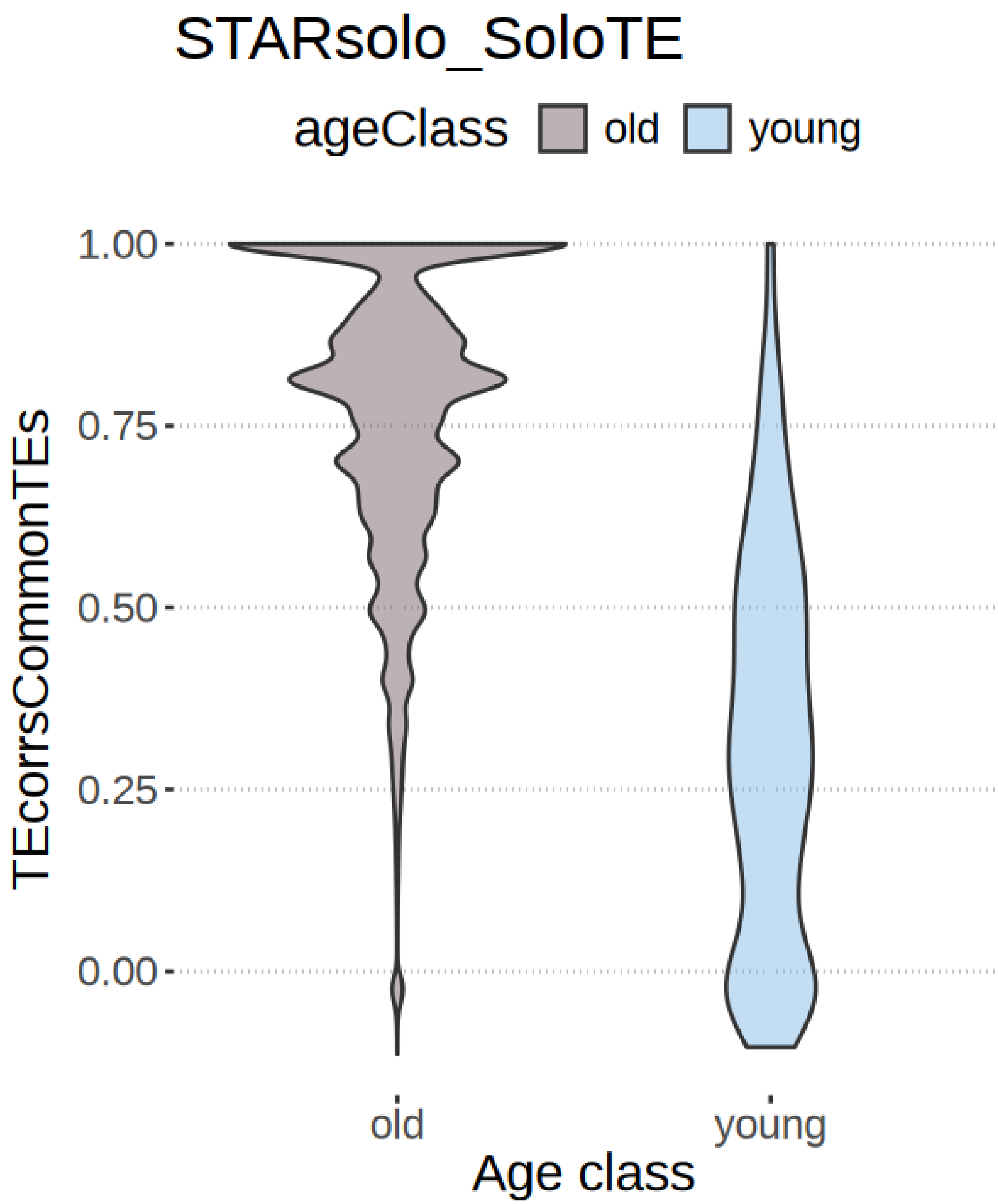

```
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_ydensity()`)."
```



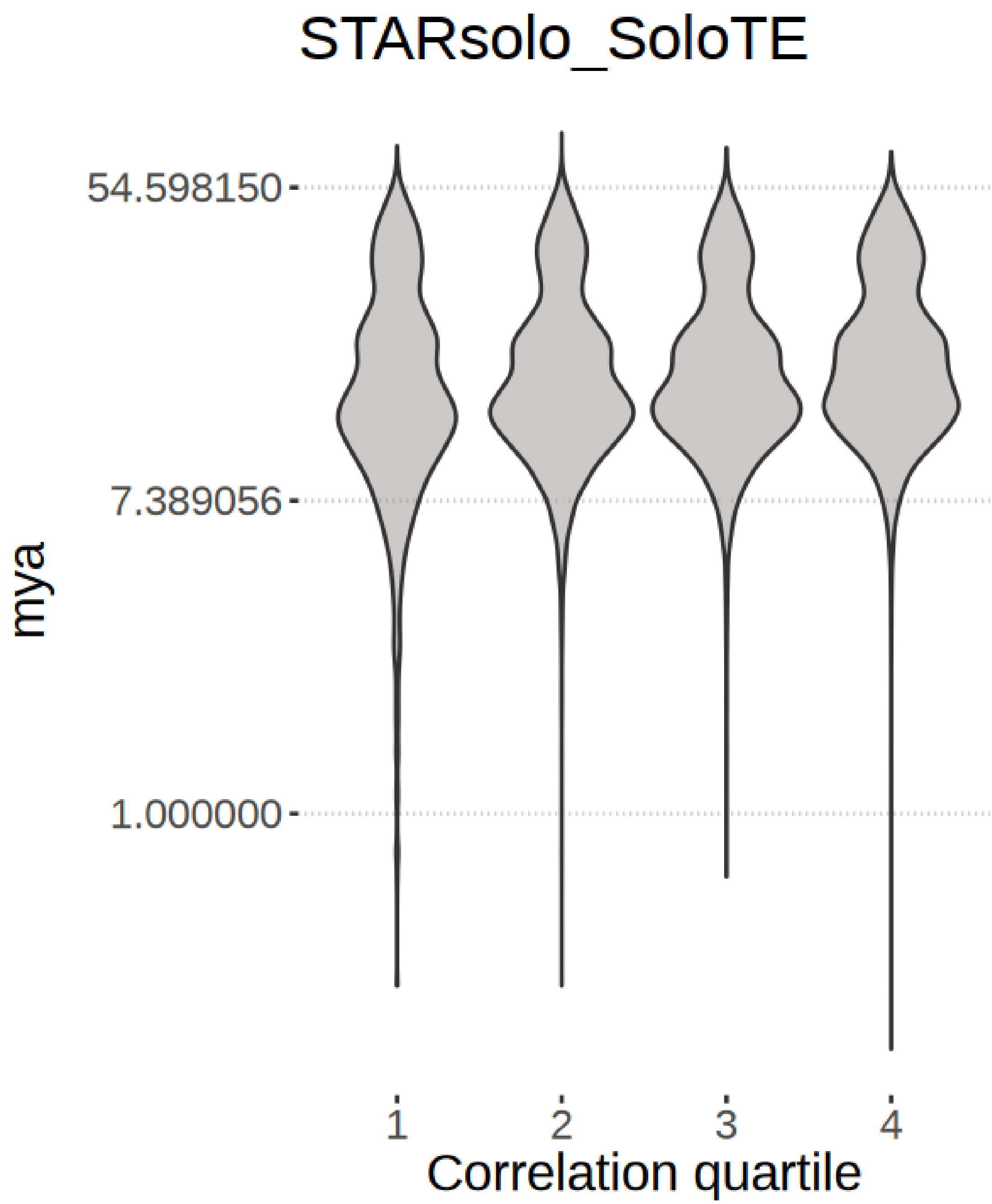

```
Warning message in geom_violin(jitter = TRUE, alpha = 0.5):
"Ignoring unknown parameters: `jitter`"
```




```
Warning message in scale_y_continuous(transform = "log"):
"log-2.718282 transformation introduced infinite values."
Warning message:
"Removed 9 rows containing non-finite outside the scale range
(`stat_ydensity()`)."
```

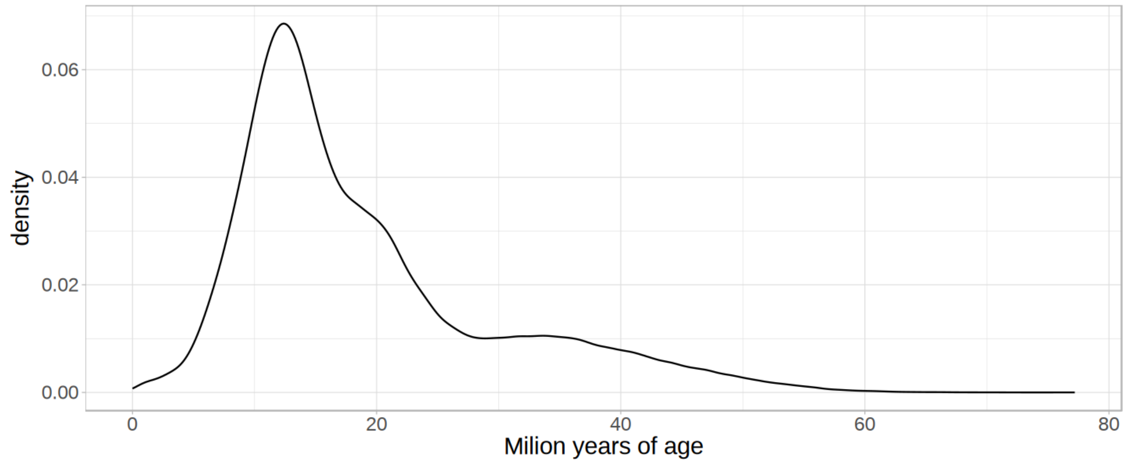


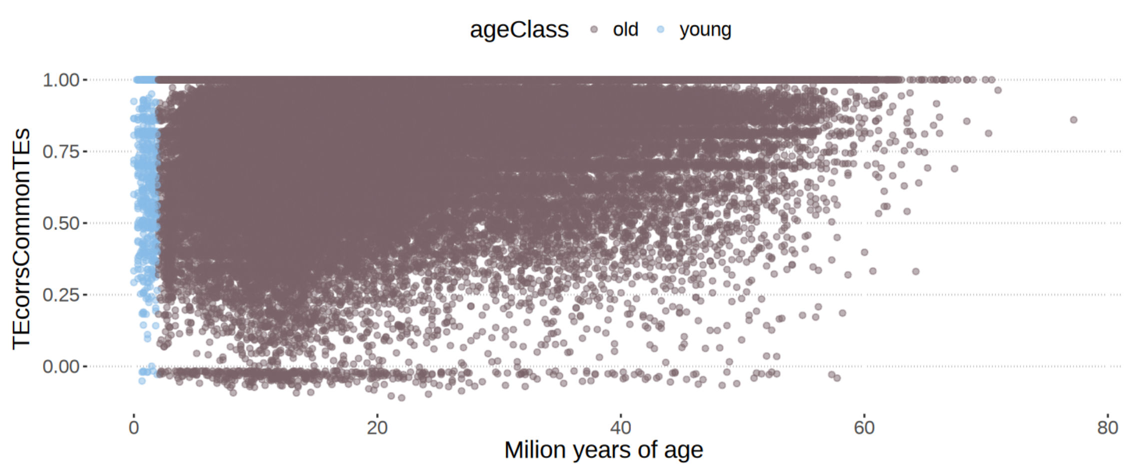
STARsolo_SoloTE
ageClass
old
young
1.00
0.75
0.50
0.25
0.00
TEcorrsCommonTEs
old
young
Age class



[12]:
```
options(repr.plot.width=12, repr.plot.height=5)

ggplot(dfsCorrelationAge[["Stellarscope_SoloTE"]], aes(x=mya)) +
  geom_density(alpha = 0.5) +
  theme_light() +
  theme(text=element_text(size=18)) +
  xlab("Milion years of age")
```

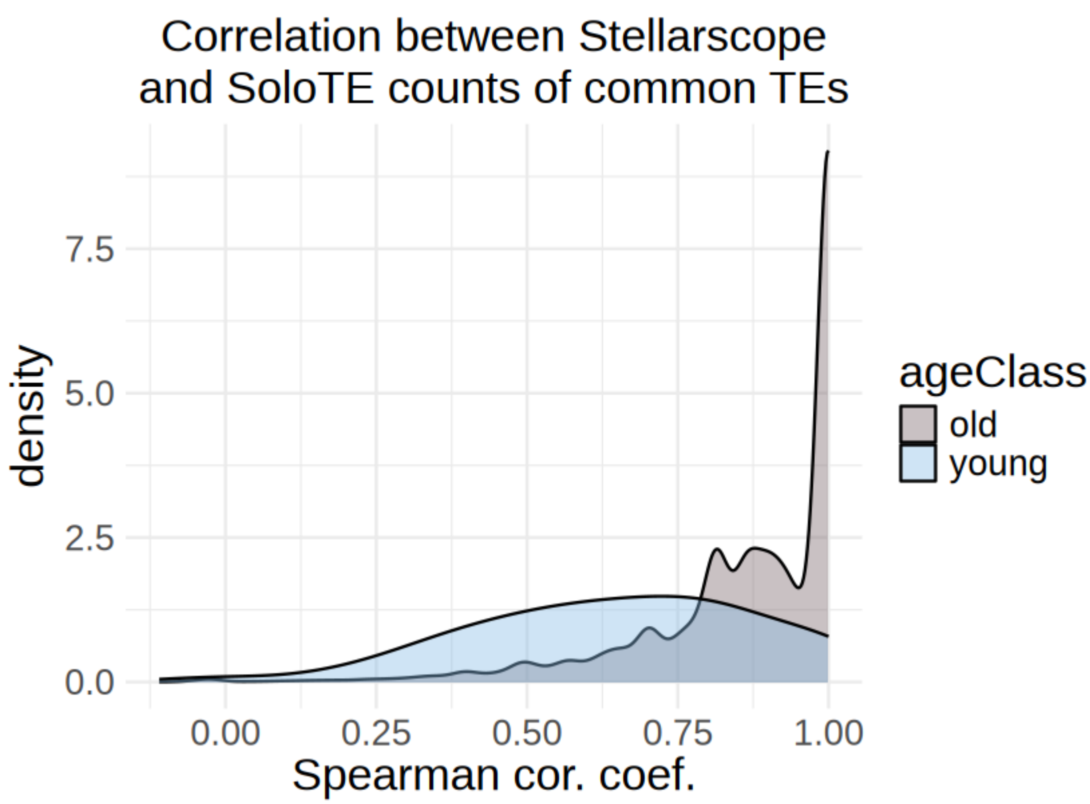
```
ggplot(dfsCorrelationAge[["Stellarscope_SoloTE"]], aes(x=mya,␣
↪y=TEcorrsCommonTEs, color=ageClass)) +
  geom_point(alpha = 0.5, ) +
  scale_color_manual(values=colorAge) +
  theme_pubclean() +
  theme(text=element_text(size=18)) +
  xlab("Milion years of age")

# this is only for the commonly detected TEs
# the pattern with tes common to SoloTE and Stellarscope was a bit different
```

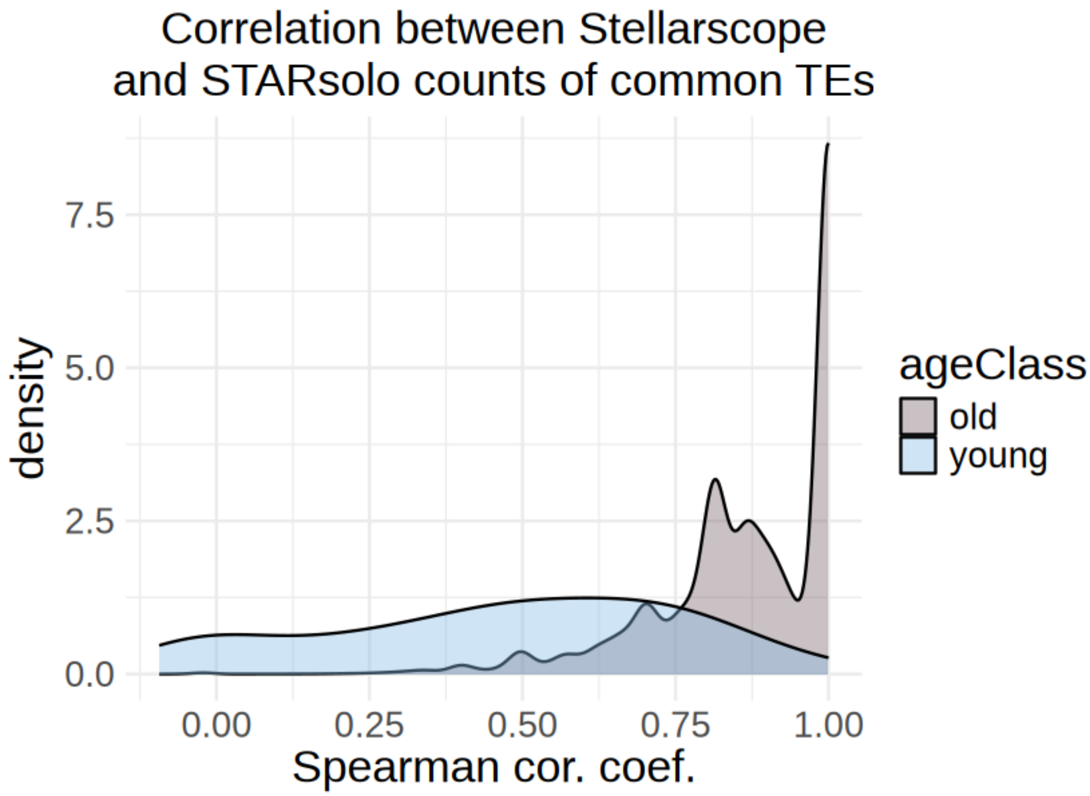


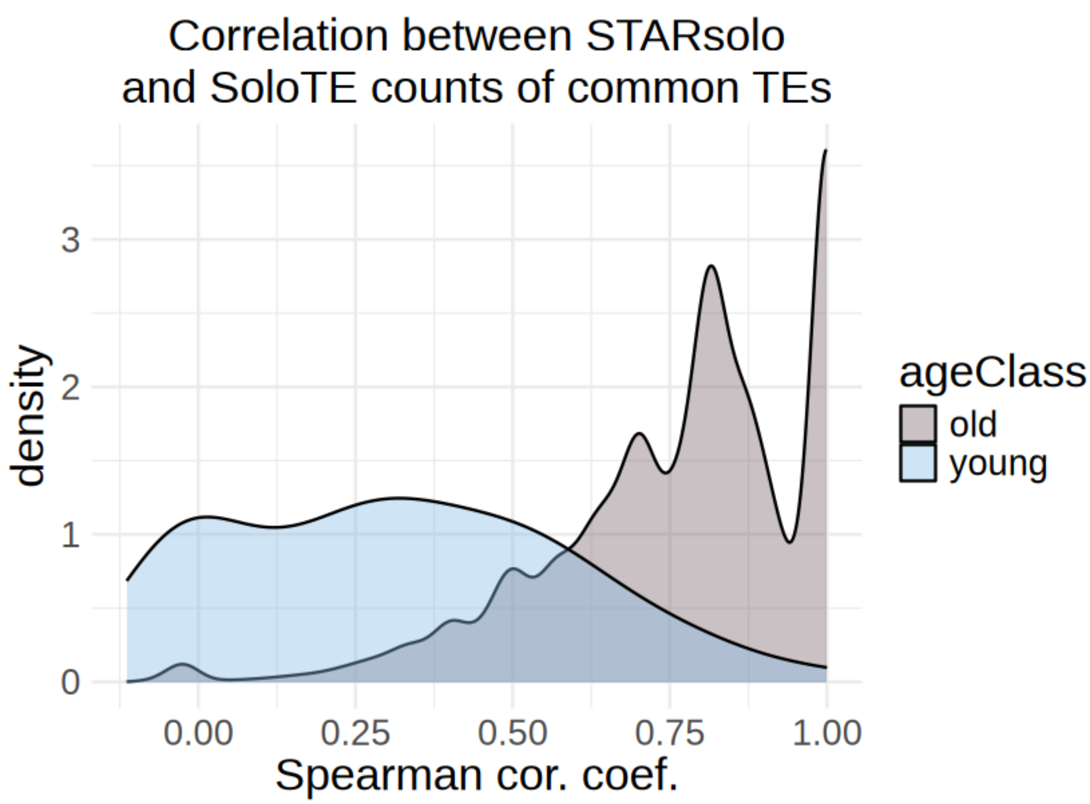


[22]:
```
options(repr.plot.width=7, repr.plot.height=5)

for(combo in names(dfsCorrelationAge)){
```

```
        tools <- unlist(strsplit(combo,"_"))

        show(ggplot(dfsCorrelationAge[[combo]]) + aes(x=TEcorrsCommonTEs,␣
↪fill=ageClass) +
                geom_density( linewidth = 0.5, alpha = 0.4, adjust = 1.5) +
                ggtitle(paste0("Correlation between ", tools[1]),
                        subtitle = paste0("and ",tools[2]," counts of common␣
↪TEs")) +
                xlab("Spearman cor. coef.") +
                scale_fill_manual(values=colorAge) +
                theme_minimal() +
                theme(text=element_text(size=20),
                        plot.title = element_text(size=20, hjust=0.5),
                        plot.subtitle = element_text(size=20, hjust=0.5))
        )
        ggsave(paste0("figures_",dataset_id,"/correlation_byAge_",combo,".
↪pdf"), width=7, height=5)

}
```




Correlation between Stellarscope
and STARsolo counts of common TEs
density
7.5
5.0
2.5
0.0
0.00
0.25
0.50
0.75
1.00
Spearman cor. coef.
ageClass
old
young


Correlation between STARsolo
and SoloTE counts of common TEs
density
3
2
1
0
0.00
0.25
0.50
0.75
1.00
Spearman cor. coef.
ageClass
old
young

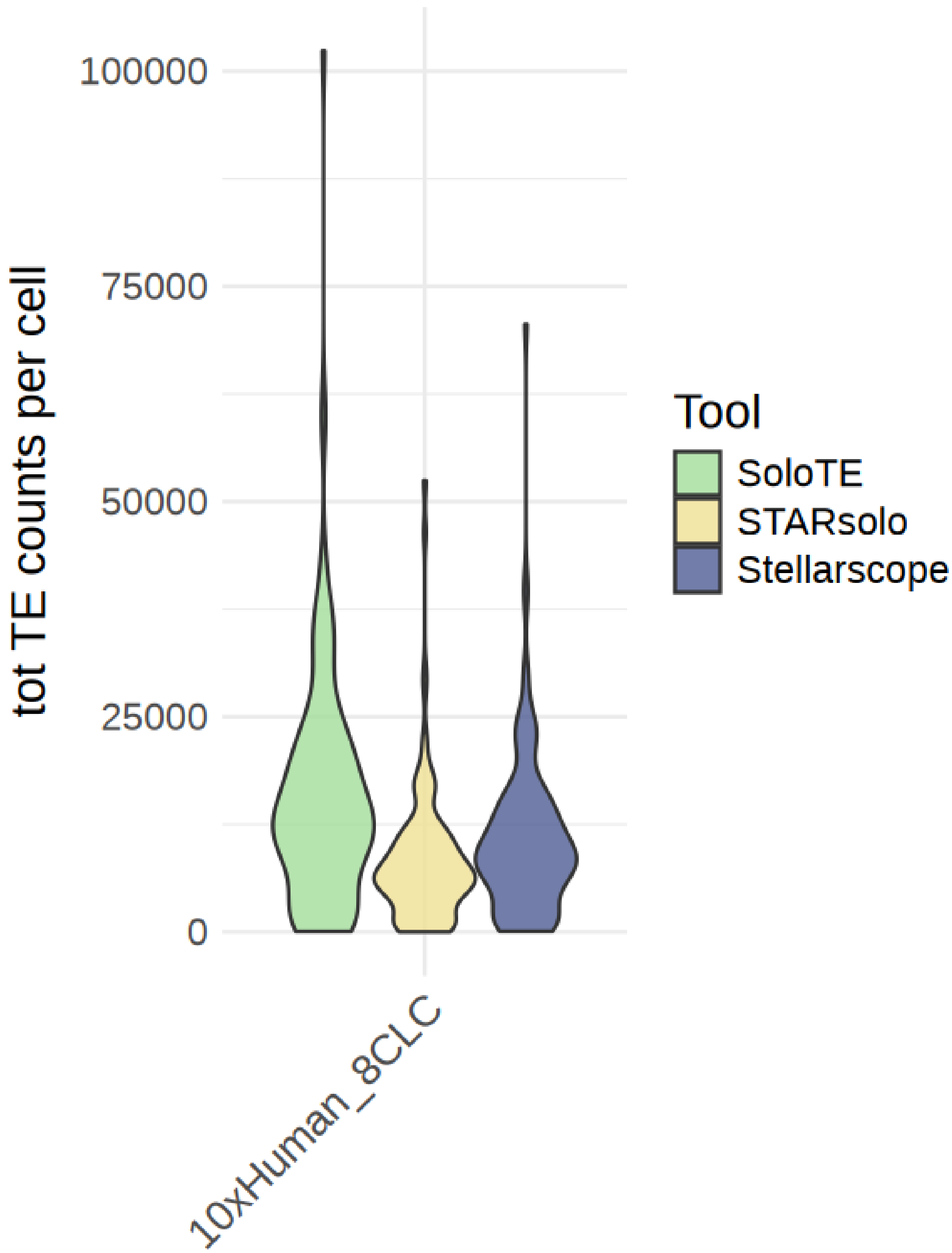
5 Correlation by family

6 Statistics on n features

[24]:
```
options(repr.plot.width=3, repr.plot.height=4)

GeomSplitViolin <- ggproto("GeomSplitViolin", GeomViolin,
                           draw_group = function(self, data, ...,␣
↪draw_quantiles = NULL) {
  data <- transform(data, xminv = x - violinwidth * (x - xmin), xmaxv = x +␣
↪violinwidth * (xmax - x))
  grp <- data[1, "group"]
  newdata <- plyr::arrange(transform(data, x = if (grp %% 2 == 1) xminv else␣
↪xmaxv), if (grp %% 2 == 1) y else -y)
  newdata <- rbind(newdata[1, ], newdata, newdata[nrow(newdata), ], newdata[1,␣
↪])
  newdata[c(1, nrow(newdata) - 1, nrow(newdata)), "x"] <- round(newdata[1, "x"])

  if (length(draw_quantiles) > 0 & !scales::zero_range(range(data$y))) {
    stopifnot(all(draw_quantiles >= 0), all(draw_quantiles <=
      1))
    quantiles <- ggplot2:::create_quantile_segment_frame(data, draw_quantiles)
    aesthetics <- data[rep(1, nrow(quantiles)), setdiff(names(data), c("x",␣
↪"y")), drop = FALSE]
    aesthetics$alpha <- rep(1, nrow(quantiles))
    both <- cbind(quantiles, aesthetics)
    quantile_grob <- GeomPath$draw_panel(both, ...)
    ggplot2:::ggname("geom_split_violin", grid::
↪grobTree(GeomPolygon$draw_panel(newdata, ...), quantile_grob))
  }
  else {
    ggplot2:::ggname("geom_split_violin", GeomPolygon$draw_panel(newdata, ...))
  }
})

geom_split_violin <- function(mapping = NULL, data = NULL, stat = "ydensity",␣
↪position = "identity", ...,
                              draw_quantiles = NULL, trim = TRUE, scale =␣
↪"area", na.rm = FALSE,
                              show.legend = NA, inherit.aes = TRUE) {
  layer(data = data, mapping = mapping, stat = stat, geom = GeomSplitViolin,
        position = position, show.legend = show.legend, inherit.aes = inherit.
↪aes,
```

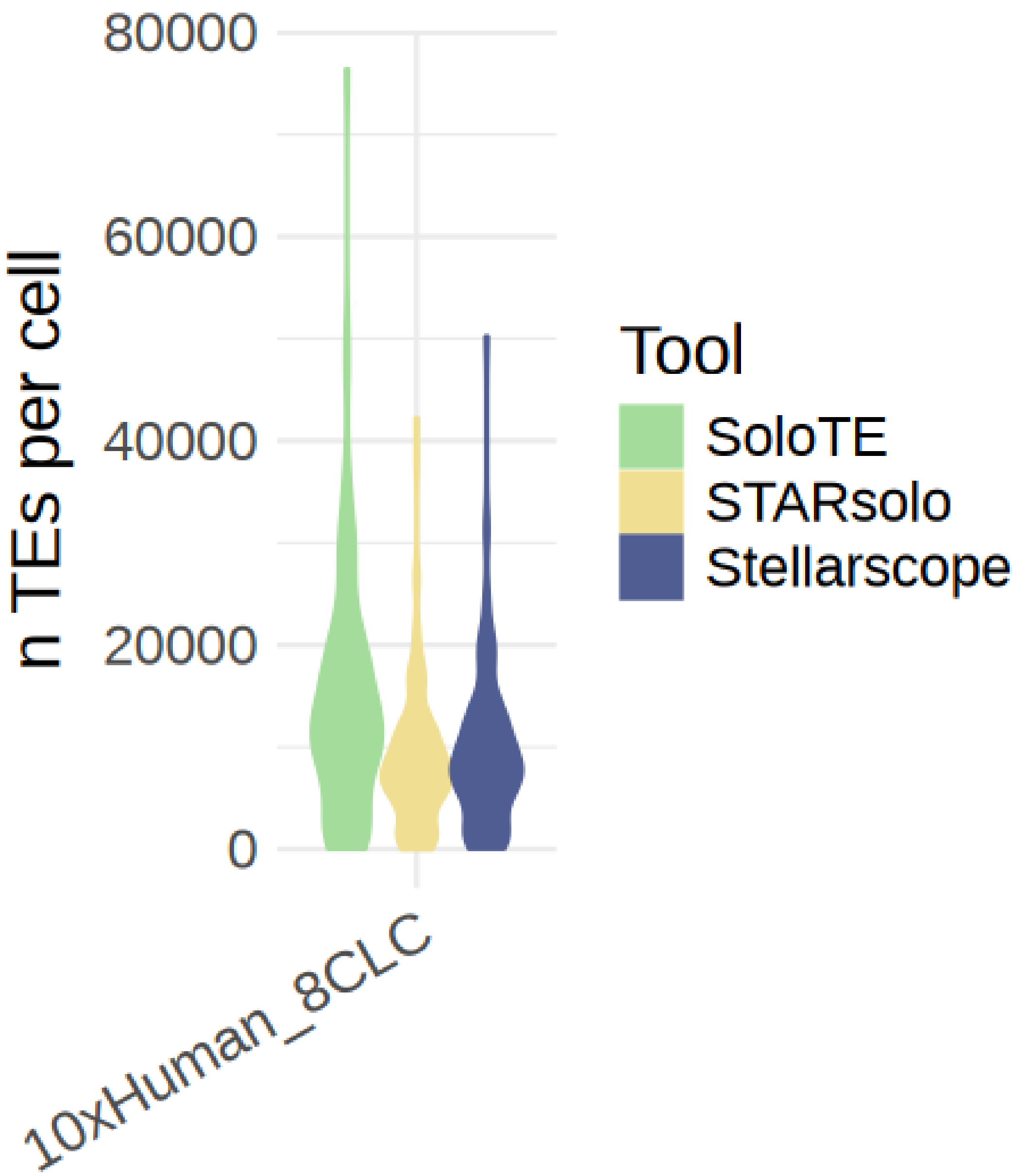
```
        params = list(trim = trim, scale = scale, draw_quantiles =
 ↪draw_quantiles, na.rm = na.rm, ...))
}
```

[25]:
```
options(repr.plot.width=5, repr.plot.height=7)

nCountsSoloTE <- objTE_soloTE[,commonCellsAll]$nCount_RNA
nCountsStellarscope <- objTE_stellarscope[,commonCellsAll]$nCount_TE
nCountsSTARsolo <- objTE_STARsolo[,commonCellsAll]$nCount_TE

df <- as.data.frame(rbind(cbind(nCountsSoloTE,"SoloTE",
 ↪sapply(strsplit(names(nCountsSoloTE), "_"),'[',1)),
                          cbind(nCountsStellarscope,"Stellarscope",
 ↪sapply(strsplit(names(nCountsStellarscope), "_"),'[',1)),
                          cbind(nCountsSTARsolo,"STARsolo",
 ↪sapply(strsplit(names(nCountsSTARsolo), "_"),'[',1))))

colnames(df) <- c("nCounts_TEs","Tool","CB")
df$Sample <- dataset_id
df$nCounts_TEs <- as.numeric(df$nCounts_TEs)

ggplot(df, aes(x=Sample, fill=Tool,  y=nCounts_TEs)) +
  geom_violin(scale = "width", alpha = 0.8, color="grey20") +
  scale_fill_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("tot TE counts per cell") +
  theme_minimal() +
  theme(text=element_text(size=17),
        plot.title = element_text(size=17, hjust=0.5),
        plot.subtitle = element_text(size=17, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/totCounts_SoloTEvsStellarscopevsSTARsolo.
 ↪pdf"), device = "pdf", width=5, height=7)
```



tot TE counts per cell
100000
75000
50000
25000
0
10xHuman_8CLC
Tool
SoloTE
STARsolo
Stellarscope

[26]:
```
options(repr.plot.width=4, repr.plot.height=5)

nFeaturesSoloTE <- objTE_soloTE[,commonCellsAll]$nFeature_RNA
nFeaturesStellarscope <- objTE_stellarscope[,commonCellsAll]$nFeature_TE
nFeaturesSTARsolo <- objTE_STARsolo[,commonCellsAll]$nFeature_TE

df <- as.data.frame(rbind(cbind(nFeaturesSoloTE,"SoloTE",␣
↪sapply(strsplit(names(nFeaturesSoloTE), "_"),'[',1)),
                          cbind(nFeaturesStellarscope,"Stellarscope",␣
↪sapply(strsplit(names(nFeaturesStellarscope), "_"),'[',1)),
                          cbind(nFeaturesSTARsolo,"STARsolo",␣
↪sapply(strsplit(names(nFeaturesSTARsolo), "_"),'[',1))))

colnames(df) <- c("nFeatures_TEs","Tool","Sample")
df$Sample <- dataset_id
df$nFeatures_TEs <- as.numeric(df$nFeatures_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))

ggplot(df, aes(x=Sample, fill=Tool, color=Tool, y=nFeatures_TEs)) +
  geom_violin(scale = "width") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("n TEs per cell") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=30, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
↪totFeatures_SoloTEvsStellarscopevsSTARsolo_violin.pdf"), device = "pdf",␣
↪width=4, height=5)
```



[27]:
```
objTE_stellarscope
mat <- GetAssayData(objTE_stellarscope, layer="counts")
sum(rowSums(mat > 0) >= 1)
```

```
An object of class Seurat
```

```
234752 features across 125 samples within 1 assay
Active assay: RNA (234752 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap
```

234752

[ ]:

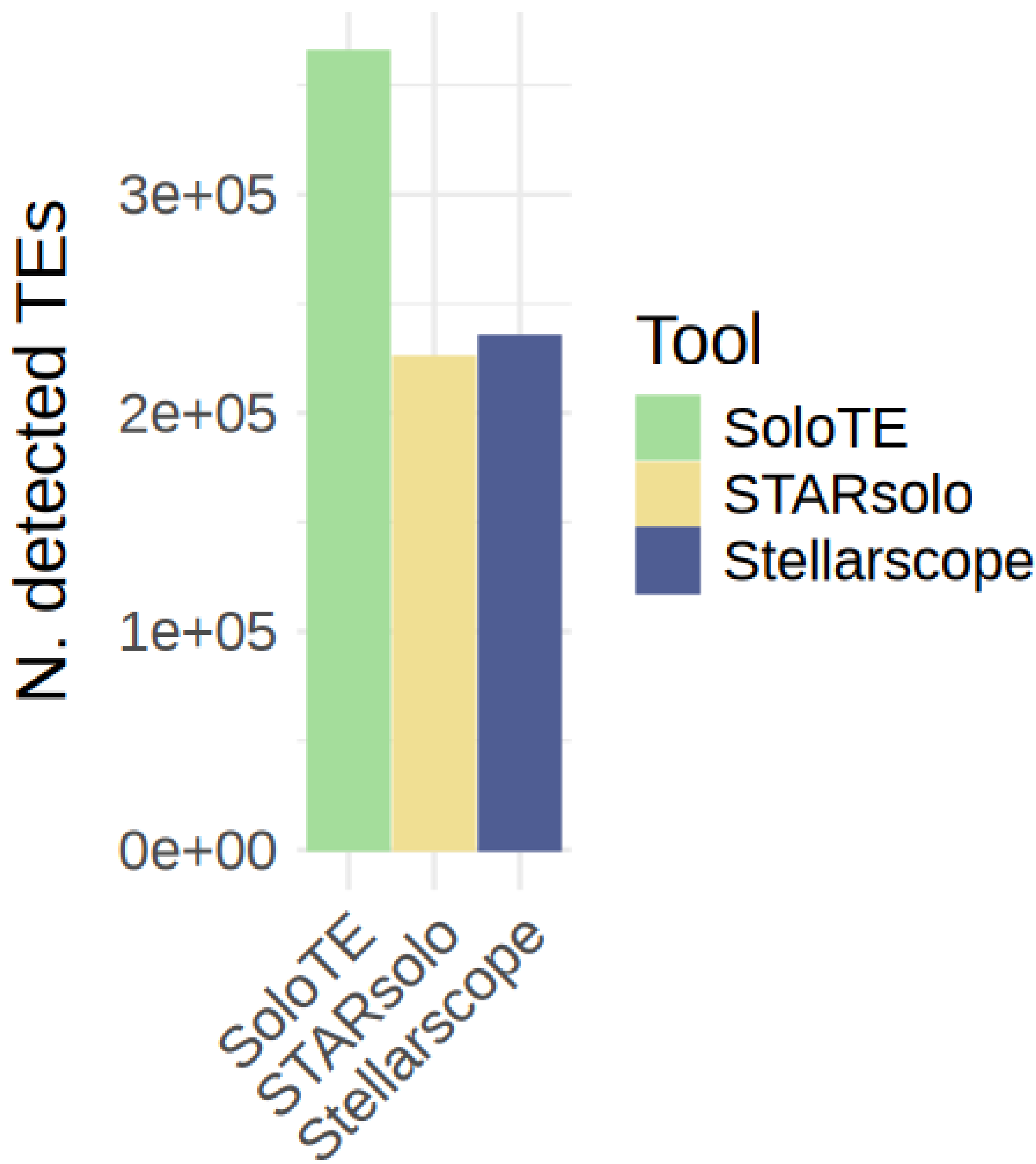
```
[28]: thrMinCells <- round(length(Cells(objTE_stellarscope))*0.02)
```

```
[29]: # save lists of TEs expressed in each sample
nTEs_soloTE <- list()
TEs_soloTE <- list()
for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],␣
 ↪layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= thrMinCells)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= thrMinCells]
}
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
 ↪ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= thrMinCells)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= thrMinCells]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident],␣
 ↪layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= thrMinCells)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= thrMinCells]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
```

```
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE), rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
             cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope), rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
             cbind(unlist(nTEs_star), names(nTEs_star), rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs","Sample","Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("N. detected TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/nTEs_SoloTEvsStellarscopevsSTARsolo_barplot.pdf"), device = "pdf", width = 4, height=5)
```

A data.frame: 3 × 3

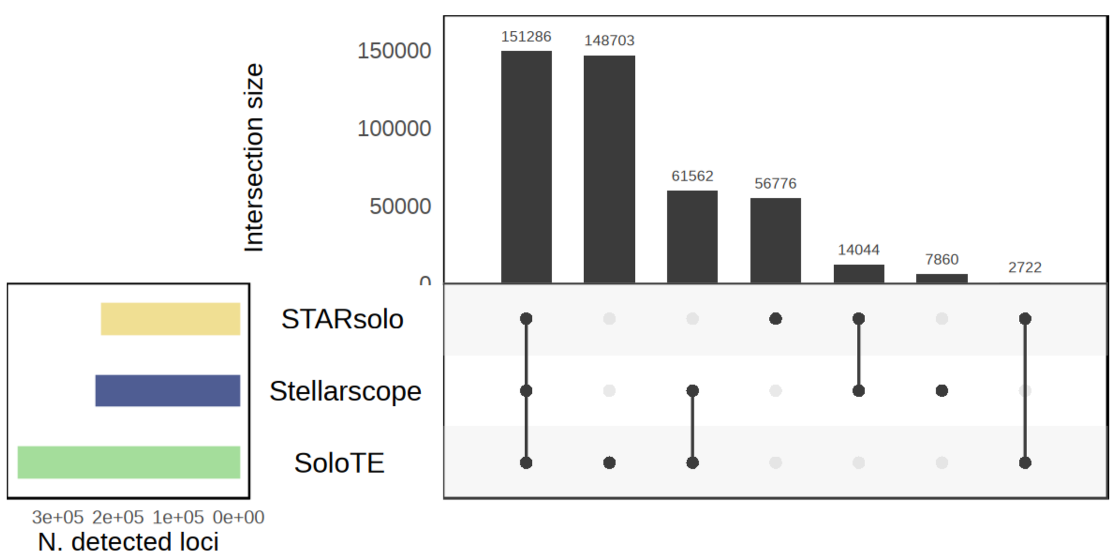
	n_TEs <dbl>	Sample <fct>	Tool <chr>
SoloTE	365333	SoloTE	SoloTE
8CLC	234752	8CLC	Stellarscope
STARsolo_TE_EM	224828	STARsolo_TE_EM	STARsolo



[30]:
```
library(UpSetR)

options(repr.plot.width=10, repr.plot.height=5)
```

```
detectedList <- list(SoloTE = tes_soloTE_conv, Stellarscope =␣
↪tes_stellarscope_conv, STARsolo = tes_starsolo_conv)

UpSetR::upset(fromList(detectedList), nintersects = 30,
              sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
              text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
              order.by=c("freq"),
              point.size=3, mb.ratio = c(0.5, 0.5),
              set_size.show=F, set_size.scale_max=␣
↪max(lengths(detectedList))+1000, #show.numbers = F,
              nsets=length(objList)+1,
              sets.x.label="N. detected loci",
              mainbar.y.label="Intersection size")
              ggsave(paste0("figures_",dataset_id,"/
↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot.pdf"), device = "pdf", width =␣
↪10, height=5)
```

```
Warning message:
"`aes_string()` was deprecated in ggplot2 3.0.0.
 Please use tidy evaluation idioms with `aes()`.
 See also `vignette("ggplot2-in-packages")` for more information.
 The deprecated feature was likely used in the UpSetR
package.
  Please report the issue to the authors."
Warning message:
"Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 Please use `linewidth` instead.
 The deprecated feature was likely used in the UpSetR
package.
  Please report the issue to the authors."
```

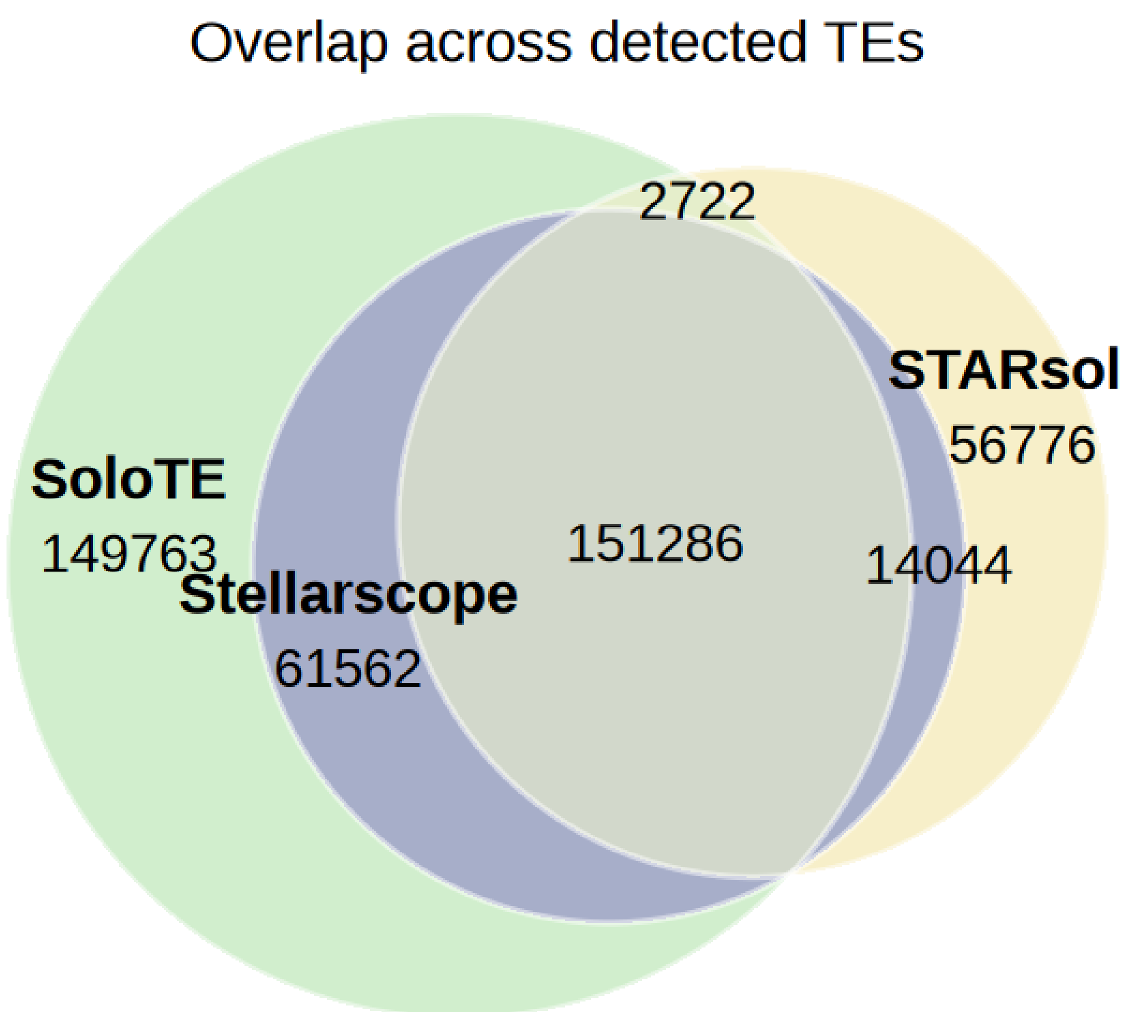


[40]:
```
options(repr.plot.width=5, repr.plot.height=5)

library(eulerr)

# remove NAs
sum(is.na(detectedList[["SoloTE"]])) # ?
detectedList[["SoloTE"]][is.na(detectedList[["SoloTE"]])] <- (1:sum(is.
  ↪na(detectedList[["SoloTE"]])))
sum(is.na(detectedList[["SoloTE"]]))

# Convert lists to sets and compute intersections
fit <- euler(detectedList)

# Plot with proportional areas
plot(fit,
     fills = colorTools,
     alpha = 0.5,
     labels = list(font = 2, cex = 1.5),       # <-- increase set label size
     quantities = list(cex = 1.4),             # <-- increase count size
     main.cex = list(cex=2),
     edges = list(col = "white", lwd = 2),
     # labels = TRUE,
     # quantities = TRUE,
     main = "Overlap across detected TEs")

pdf(paste0("figures_", dataset_id, "/overlap_euler_withSTAR.pdf"), width=5,␣
  ↪height=5)
plot(fit,
     fills = colorTools,
     alpha = 0.5,
     labels = list(font = 2, cex = 1.5),       # <-- increase set label size
     quantities = list(cex = 1.4),             # <-- increase count size
     main.cex = list(cex=2),
     edges = list(col = "white", lwd = 2),
     # labels = TRUE,
     # quantities = TRUE,
     main = "Overlap across detected TEs")
dev.off()
```

0

0

agg_record_1158971242: 2



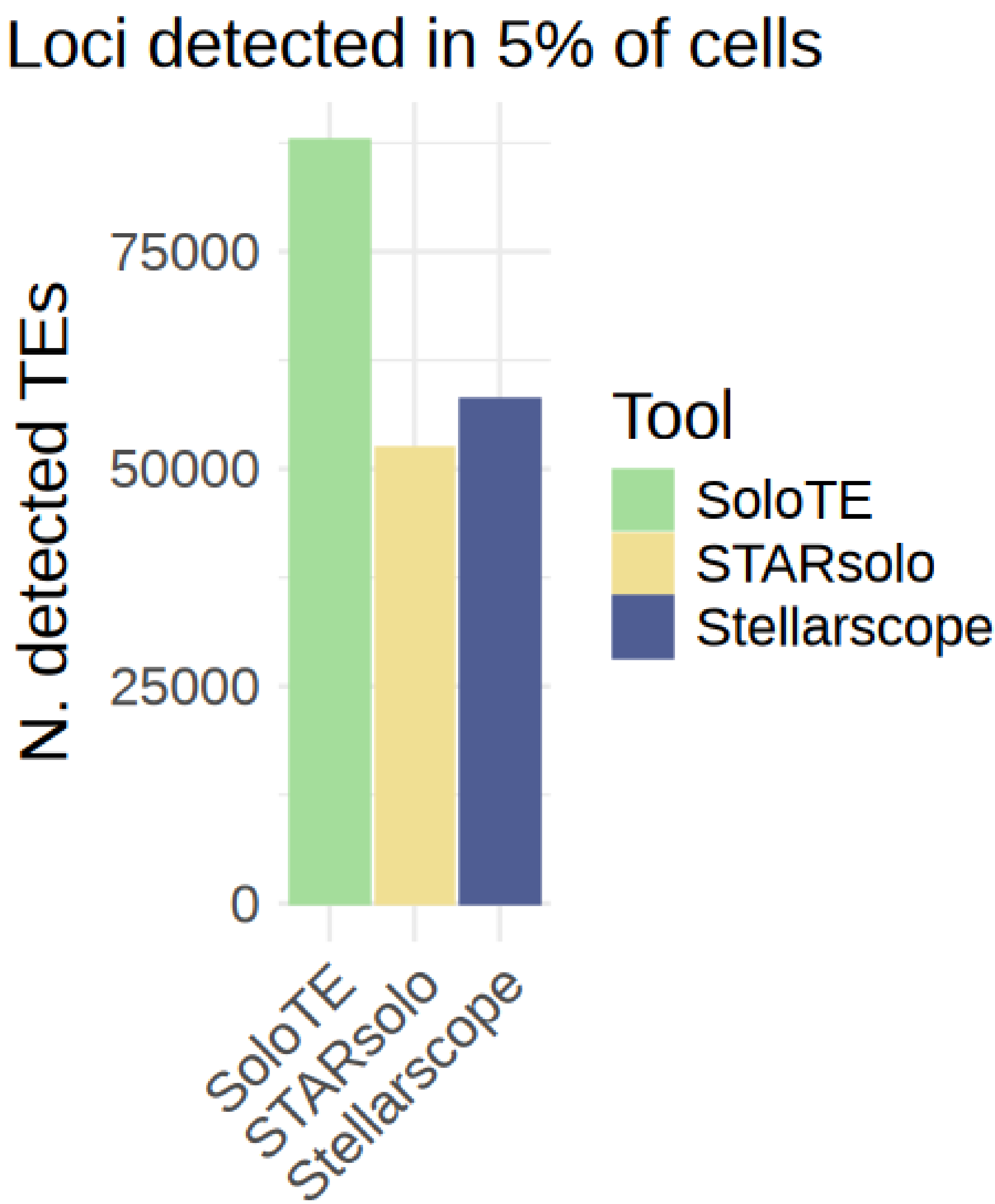
[41]:
```
# Number of TEs detected above a certain threshold of cells ( 5% of the dataset)

cellNum <- round(length(Cells(objTE_soloTE))*0.05)
# save lists of TEs expressed in each sample
nTEs_soloTE <- list()
TEs_soloTE <- list()

for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],
 ↪layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
```

```r
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident], layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE), rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
             cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope), rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
             cbind(unlist(nTEs_star), names(nTEs_star), rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs","Sample","Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle(paste0("Loci detected in 5% of cells")) +
  xlab("") +
  ylab("N. detected TEs") +
```

```
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
 ↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_thr5percCells.pdf"), device =␣
 ↪"pdf", width = 5, height=5)
```

A data.frame: 3 × 3

	n_TEs	Sample	Tool
	<dbl>	<fct>	<chr>
SoloTE	87816	SoloTE	SoloTE
8CLC	57998	8CLC	Stellarscope
STARsolo_TE_EM	52314	STARsolo_TE_EM	STARsolo



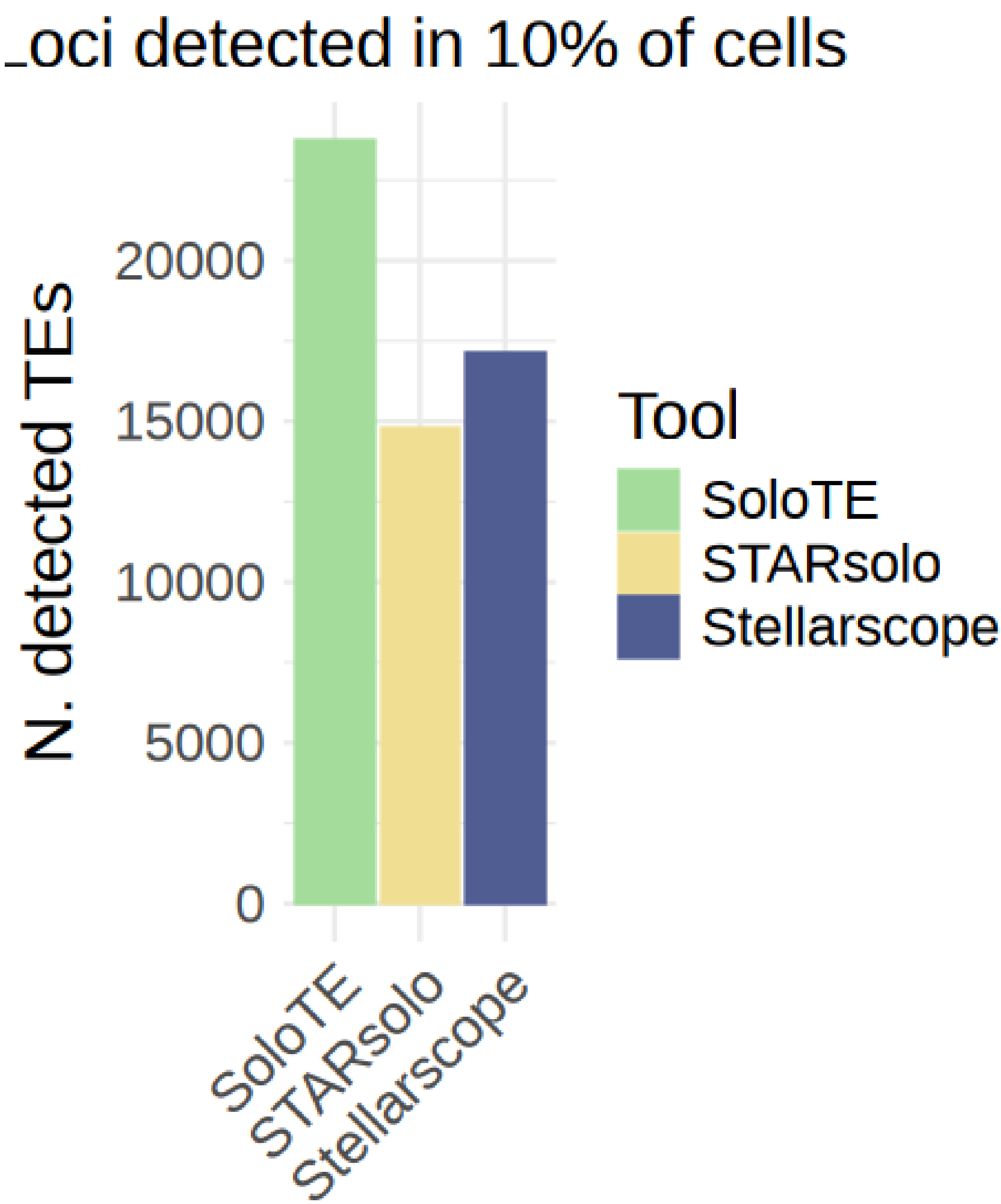
[42]:
```
# Number of TEs detected above a certain threshold of cells ( 10% of the
 ↪dataset)

cellNum <- round(length(Cells(objTE_soloTE))*0.1)
# save lists of TEs expressed in each sample
```

```r
nTEs_soloTE <- list()
TEs_soloTE <- list()

for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident], 
  ↪layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
  ↪ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident], 
  ↪layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE), 
  ↪rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
             cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope), 
  ↪rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
             cbind(unlist(nTEs_star), names(nTEs_star), 
  ↪rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs","Sample","Tool")
df <- as.data.frame(df)
```

```
df$n_TEs <- as.numeric(df$n_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle(paste0("Loci detected in 10% of cells")) +
  xlab("") +
  ylab("N. detected TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_thr10percCells.pdf"), device =␣
↪"pdf", width = 5, height=5)
```

A data.frame: 3 × 3

	n_TEs <dbl>	Sample <fct>	Tool <chr>
SoloTE	23742	SoloTE	SoloTE
8CLC	17125	8CLC	Stellarscope
STARsolo_TE_EM	14753	STARsolo_TE_EM	STARsolo



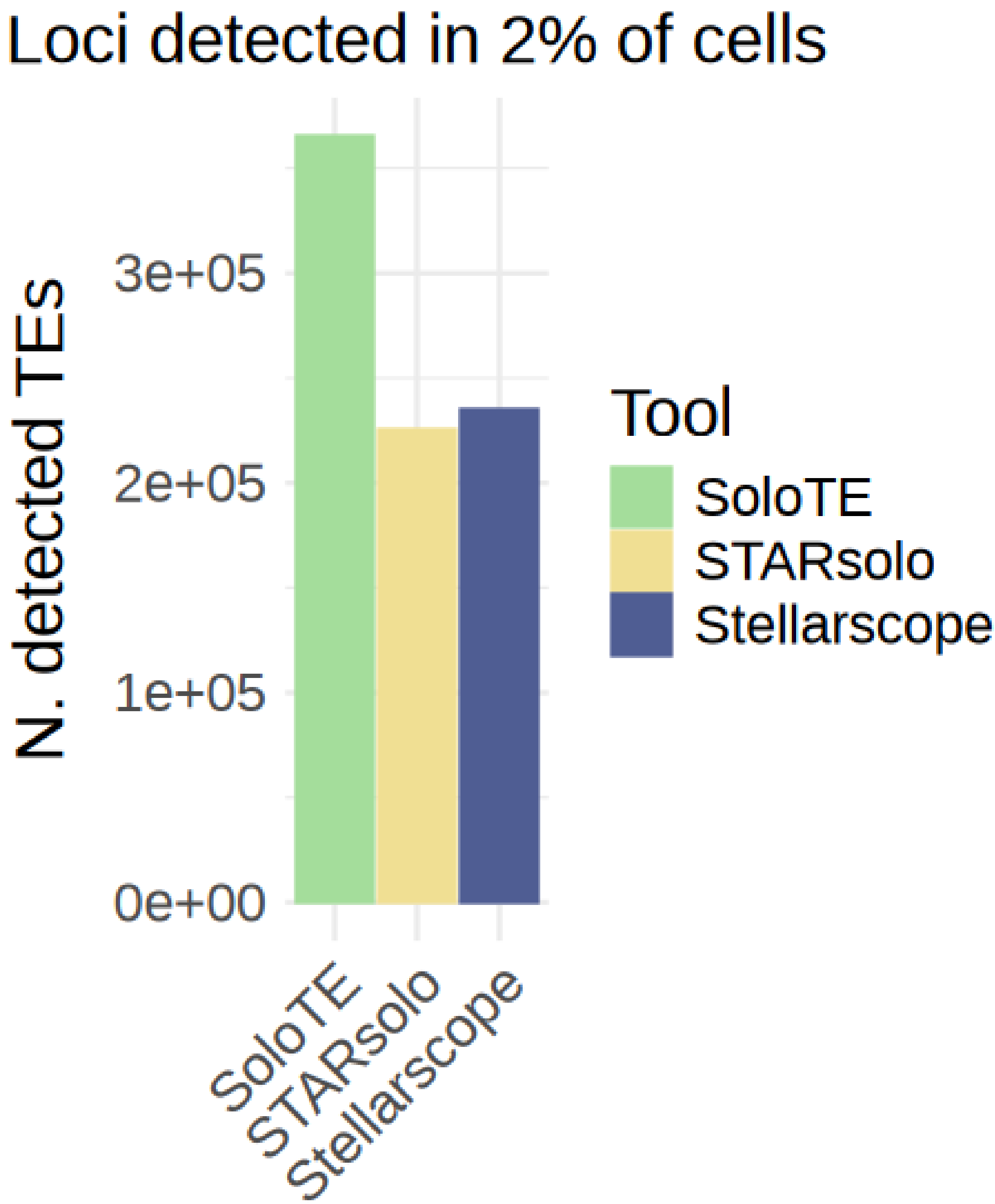
[43]:
```
# Number of TEs detected above a certain threshold of cells ( 2% of the dataset)

cellNum <- round(length(Cells(objTE_soloTE))*0.02)
# save lists of TEs expressed in each sample
nTEs_soloTE <- list()
```

```r
TEs_soloTE <- list()

for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident], layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident], layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE), rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
             cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope), rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
             cbind(unlist(nTEs_star), names(nTEs_star), rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs","Sample","Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)
```

```
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle(paste0("Loci detected in 2% of cells")) +
  xlab("") +
  ylab("N. detected TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
 ↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_thr2percCells.pdf"), device =␣
 ↪"pdf", width = 5, height=5)
```

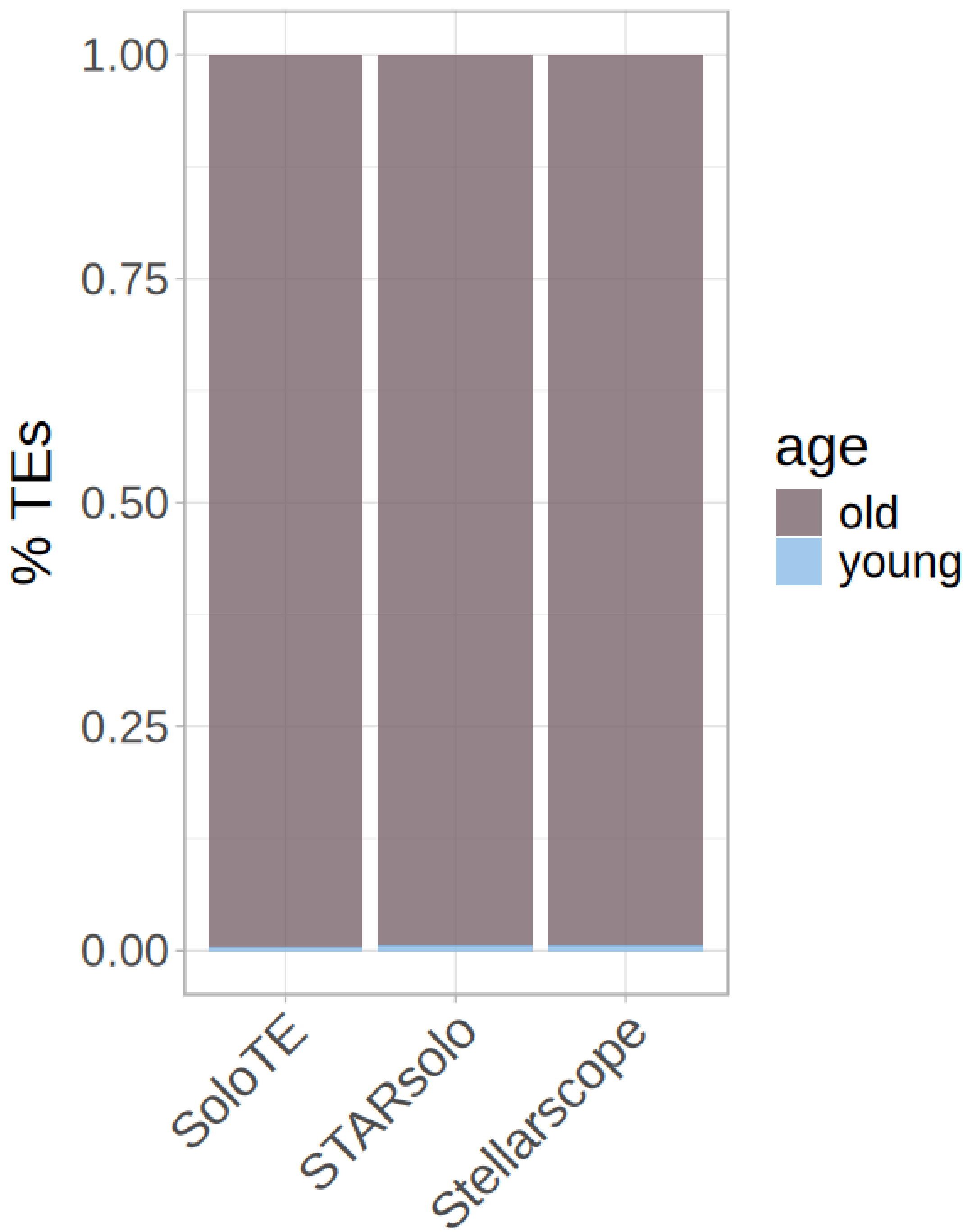
A data.frame: 3 × 3

	n_TEs	Sample	Tool
	<dbl>	<fct>	<chr>
SoloTE	365333	SoloTE	SoloTE
8CLC	234752	8CLC	Stellarscope
STARsolo_TE_EM	224828	STARsolo_TE_EM	STARsolo



[44]:
```
options(repr.plot.width=5, repr.plot.height=7)
TEs_soloTE_df$age <- conversionTable[match(TEs_soloTE_df$locus,␣
↪conversionTable$soloteID),"ageClass"]
TEs_stellarscope_df$age <- conversionTable[match(TEs_stellarscope_df$locus,␣
↪conversionTable$stellarscopeID),"ageClass"]
```

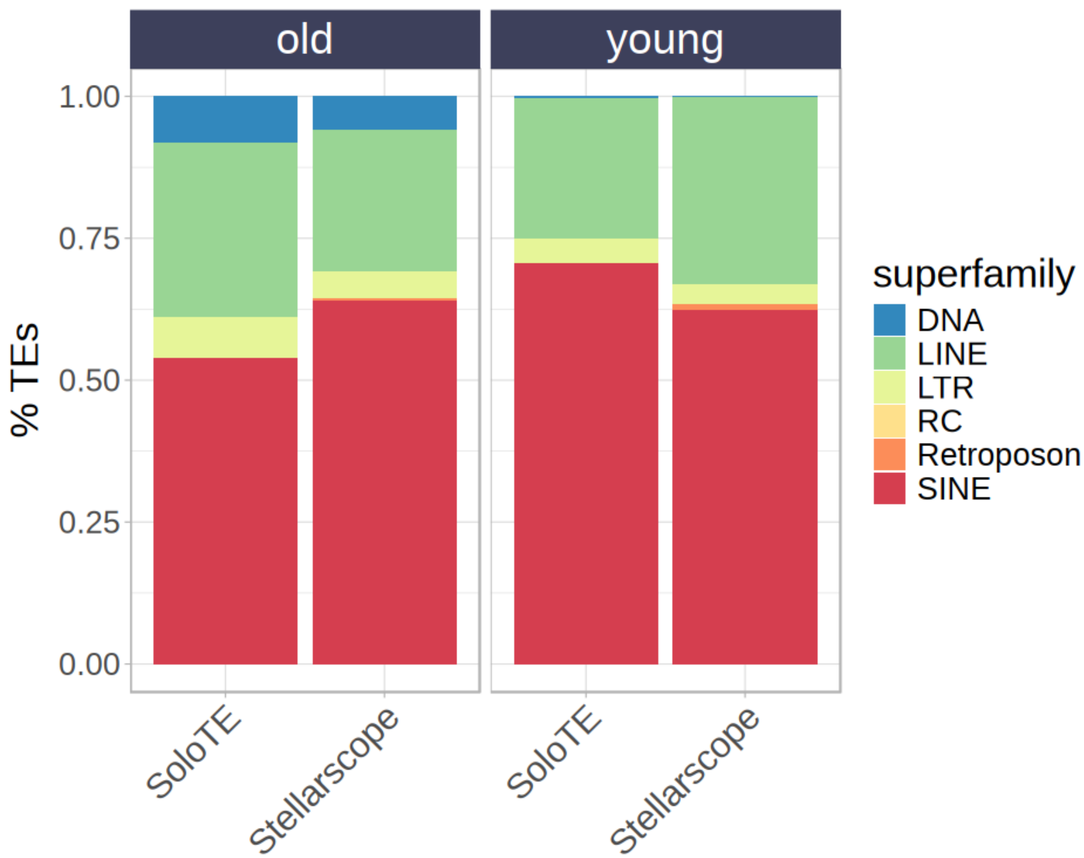
```
TEs_star_df$age <- conversionTable[match(TEs_star_df$locus, conversionTable$stellarscopeID),"ageClass"]

TEs_soloTE_df$tool <- "SoloTE"
TEs_stellarscope_df$tool <- "Stellarscope"
TEs_star_df$tool <- "STARsolo"
df <- rbind(TEs_soloTE_df,TEs_stellarscope_df,TEs_star_df)

df <- df[!is.na(df$age),]

ggplot(df, aes(x=tool, fill=age))+
  geom_bar(position="fill", alpha = 0.8) +
  scale_fill_manual(values=colorAge) +
  ggtitle("") +
  xlab("") +
  ylab("% TEs") +
  theme_light() +
  theme(text=element_text(size=20),
        strip.text.x = element_text(size=22, hjust=0.5),
        strip.background = element_rect(fill="#7E7E7E"),
        axis.text.x = element_text(size=18, angle=45, hjust=1))

ggsave(paste0("figures_",dataset_id,"/nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_age_tool_celltype.pdf"), device = "pdf", width=5, height=7)
```


1.00
0.75
0.50
0.25
0.00
% TEs
SoloTE
STARsolo
Stellarscope
age
old
young

```
[45]: options(repr.plot.width=8, repr.plot.height=7)
soloTEageFamily <- table(conversionTable[match(Features(objTE_soloTE),␣
 ↪conversionTable$soloteID),"ageClass"],
                          conversionTable[match(Features(objTE_soloTE),␣
 ↪conversionTable$soloteID),"class"])
soloTEageFamilyOld <-␣
 ↪cbind(soloTEageFamily["old",],colnames(soloTEageFamily),"old","SoloTE")
colnames(soloTEageFamilyOld) <- c("nTEs","superfamily","age","tool")
soloTEageFamilyYoung <-␣
 ↪cbind(soloTEageFamily["young",],colnames(soloTEageFamily),"young","SoloTE")
colnames(soloTEageFamilyYoung) <- c("nTEs","superfamily","age","tool")
soloTEageFamily <- rbind(soloTEageFamilyOld, soloTEageFamilyYoung)
soloTEageFamily <- as.data.frame(soloTEageFamily)
soloTEageFamily$nTEs <- as.numeric(soloTEageFamily$nTEs)

stellarscopeageFamily <-␣
 ↪table(conversionTable[match(Features(objTE_stellarscope),␣
 ↪conversionTable$stellarscopeID),"ageClass"],
                          conversionTable[match(Features(objTE_stellarscope),␣
 ↪conversionTable$stellarscopeID),"class"])
stellarscopeageFamilyOld <-␣
 ↪cbind(stellarscopeageFamily["old",],colnames(stellarscopeageFamily),"old","Stellarscope")
colnames(stellarscopeageFamilyOld) <- c("nTEs","superfamily","age","tool")
stellarscopeageFamilyYoung <-␣
 ↪cbind(stellarscopeageFamily["young",],colnames(stellarscopeageFamily),"young","Stellarscope
colnames(stellarscopeageFamilyYoung) <- c("nTEs","superfamily","age","tool")
stellarscopeageFamily <- rbind(stellarscopeageFamilyOld,␣
 ↪stellarscopeageFamilyYoung)
stellarscopeageFamily <- as.data.frame(stellarscopeageFamily)
stellarscopeageFamily$nTEs <- as.numeric(stellarscopeageFamily$nTEs)


ageFamilyDf <- rbind(soloTEageFamily, stellarscopeageFamily)
ageFamilyDf$superfamily <- gsub(ageFamilyDf$superfamily, pattern="\\?",␣
 ↪replacement="")

ageFamilyDf <- ageFamilyDf[ageFamilyDf$nTEs>0,]


ggplot(ageFamilyDf, aes(x=tool, fill=superfamily, y=nTEs)) +
  facet_wrap(~age) +
  geom_col(position="fill") +
 ␣
 ↪#scale_fill_manual(values=adjustcolor(c("#5E50A1","#3787BC","#67C1A3","#AADCA2","#E6F596"),
 ↪alpha.f = 0.8)) +
  scale_fill_brewer(palette = "Spectral", direction = -1)+
```

```
  #palette = "Spectral", direction = -1)+
  # scale_color_manual(values=c("#81B29A","#9181A6"))+
  ggtitle("") +
  xlab("") +
  ylab("% TEs") +
  theme_light() +
  theme(text=element_text(size=20),
        strip.text.x = element_text(size=22, hjust=0.5),
        strip.background = element_rect(fill="#3d405b"),   #"#e78063"
        axis.text.x = element_text(size=18, angle=45, hjust=1))

ggsave(paste0("figures_",dataset_id,"/
  ↪nTEs_SoloTEvsStellarscope_barplot_age_order.pdf"), device = "pdf")
```

```
Saving 6.67 x 6.67 in image
```

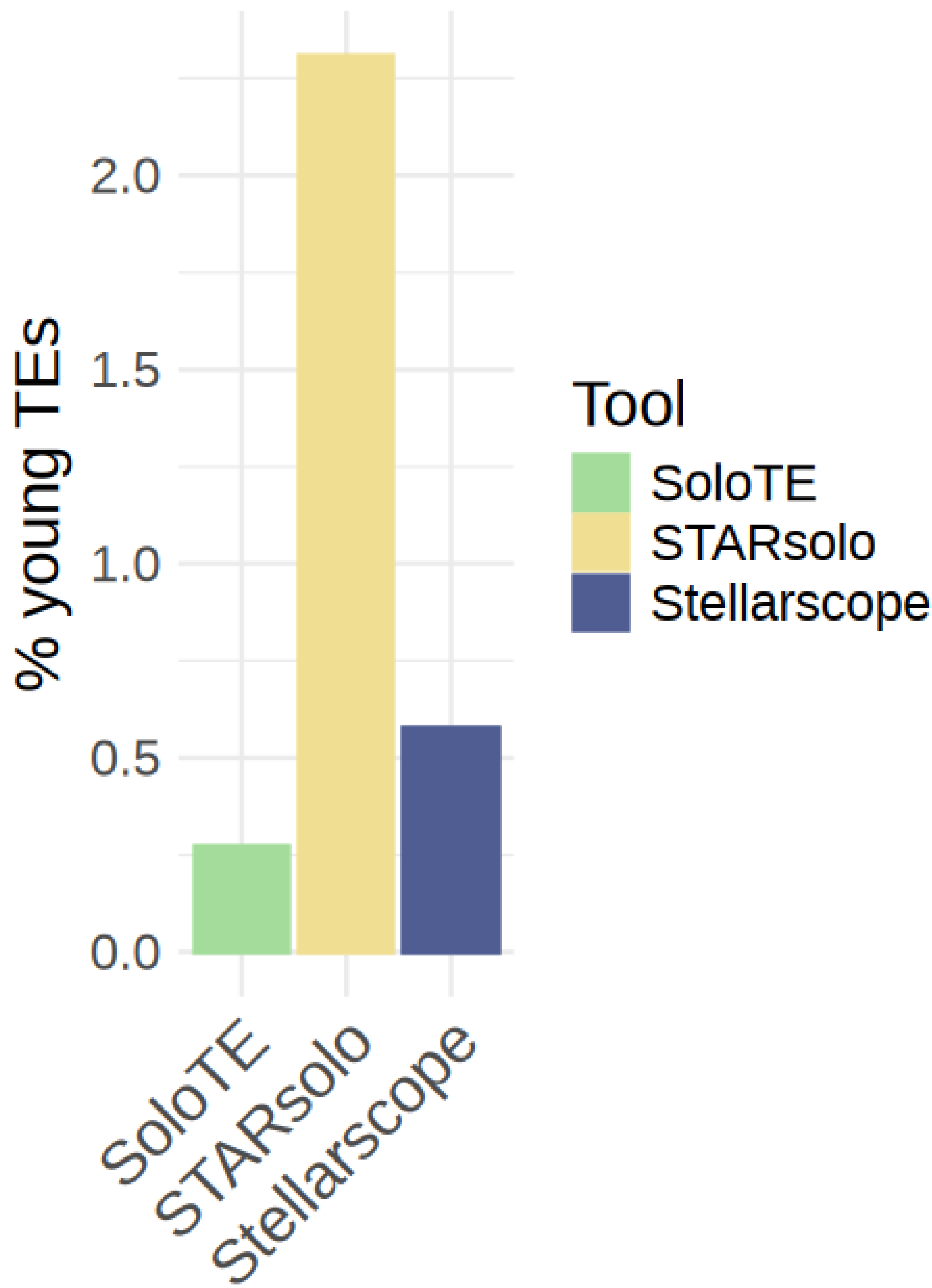


[46]:

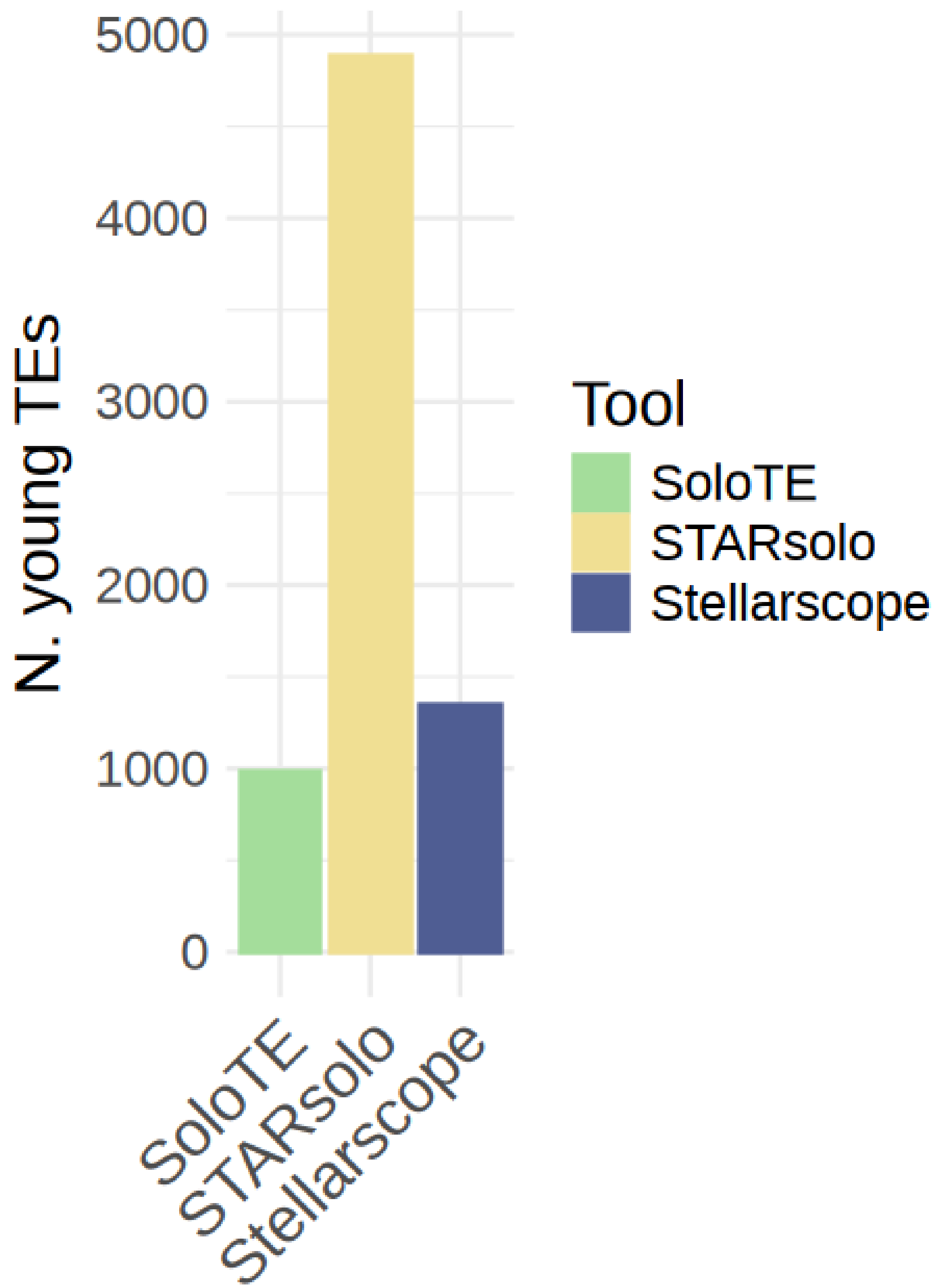
```
options(repr.plot.width=4, repr.plot.height=6)

nOldSoloTE <- table(conversionTable[match(Features(objTE_soloTE),␣
 ↪conversionTable$soloteID),"ageClass"])["old"]
nYoungSoloTE <- table(conversionTable[match(Features(objTE_soloTE),␣
 ↪conversionTable$soloteID),"ageClass"])["young"]
nOldStellarscope <- table(conversionTable[match(Features(objTE_stellarscope),␣
 ↪conversionTable$stellarscopeID),"ageClass"])["old"]
nYoungStellarscope <- table(conversionTable[match(Features(objTE_stellarscope),␣
 ↪conversionTable$stellarscopeID),"ageClass"])["young"]
nOldSTARsolo <- table(conversionTable[match(Features(objTE_STARsolo),␣
 ↪conversionTable$stellarscopeID),"ageClass"])["old"]
nYoungSTARsolo <- table(conversionTable[match(Features(objTE_STARsolo),␣
 ↪conversionTable$stellarscopeID),"ageClass"])["young"]


table(conversionTable[match(Features(objTE_stellarscope),␣
 ↪conversionTable$stellarscopeID),"ageClass"])

df <- rbind( c(nOldSoloTE , nOldSoloTE / (nOldSoloTE + nYoungSoloTE), "old",␣
 ↪"SoloTE"),
             c(nYoungSoloTE, nYoungSoloTE / (nOldSoloTE + nYoungSoloTE),␣
 ↪"young", "SoloTE"),
             c(nOldStellarscope, nOldStellarscope / (nOldStellarscope +␣
 ↪nYoungStellarscope), "old", "Stellarscope"),
             c(nYoungStellarscope, nYoungStellarscope  / (nOldStellarscope +␣
 ↪nYoungStellarscope), "young", "Stellarscope"),
             c(nOldSTARsolo, nOldSTARsolo / (nOldSTARsolo + nYoungSTARsolo),␣
 ↪"old", "STARsolo"),
             c(nYoungSTARsolo, nYoungSTARsolo  / (nOldSTARsolo +␣
 ↪nYoungSTARsolo), "young", "STARsolo")
             )
df <- as.data.frame(df)


colnames(df) <- c("n_TEs","n_TEs_prop", "AgeClass", "Tool")
df$n_TEs <- as.numeric(df$n_TEs)
df$n_TEs_prop <- as.numeric(df$n_TEs_prop)
df$n_TEs_perc <- as.numeric(df$n_TEs_prop) * 100



ggplot(df[df$AgeClass=="young",], aes(x=Tool, fill=Tool, color=Tool,␣
 ↪y=n_TEs_perc)) +
  geom_col(position = "stack") +
  scale_fill_manual(values=colorTools)+
```

```
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("% young TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=18, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/barplot_percYoungTEs.pdf"), device =
 "pdf", width = 4, height=6)

ggplot(df[df$AgeClass=="young",], aes(x=Tool, fill=Tool, color=Tool, y=n_TEs))
 +
  geom_col(position = "stack") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("N. young TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=18, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/barplot_nYoungTEs.pdf"), device = "pdf",
 width = 4, height=6)
```

```
   old  young
232358   1344
```



% young TEs
2.0
1.5
1.0
0.5
0.0
SoloTE
STARsolo
Stellarscope
Tool
SoloTE
STARsolo
Stellarscope


N. young TEs
5000
4000
3000
2000
1000
0
SoloTE
STARsolo
Stellarscope
Tool
SoloTE
STARsolo
Stellarscope